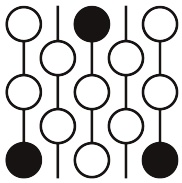
Review of Problem 2 – Siege



First Proof Third Batch
Reviewer 1
October 8–9, 2026

1 Review

The paper gives the correct solution of the requested minimax rate (up to log factors) using novel ideas for estimation and lower bound alternatives, compared to the published literature. The estimation relies on a plugin approach for the explicit reconstruction and detailed arguments to derive the correct convergence rate. The error decomposition is different from the human proof and simplifies some of their upper bounds. The lower bound relies on spherical perturbations in the functional class, i.e. the domain of the underlying nonlinear forward operator and uses Assouad cube techniques, which differs from the human and most AI proofs.

Correctness. The stated result is correct. For the upper bound, the construction of the estimator, the error decomposition and the treatment of the different error terms are novel and generally correct. Some arguments would require more details, but could be worked out quickly, as it seems. The only clear mathematical mistake is in Section 7.1: the notation $\partial^m \chi_k$ for a multivariate function χ_k and integer m, where partial derivatives with multi-indices m are expected. The lower bound proof is innovative with its forward estimate to control the KL divergence, but seems correct as well. All proofs are very dense and even a reviewer working in the field has difficulties to check precisely without too much own work. The overall assessment is that there are minor gaps which seem relatively easy to fill.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5

Exposition. The human language presentation is not well done, putting short phrases, short arguments and inline formulas densely together. Section 6 lists standard facts without introducing certain notation or the meaning of the statements. A good point are references to equation numbers and general properties as well as Section 5 with an overview. Overall, the exposition is ok, but requires substantial efforts to follow and especially to verify.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions. The paper lists used tools with correct references. Some results, e.g. Lemma 8.1, 8.3, 8.5(4) are stated as is or very similarly in the literature. This should be cited and in case of differences discussed. There is no reference to prior (suboptimal) solutions of the problem (exist for the upper bound) or related approaches.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

3 The problem

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, be the closed Euclidean unit ball. For $f \in C^\alpha(D)$ with Hölder–Zygmund smoothness $\alpha > 1$ (and f bounded away from 0), let $u = u_f$ be the unique solution of Darcy's problem

$$\operatorname{div}(f\nabla u) = 1 \ \text{ on } D, \qquad u|_{\partial D} = 0.$$

We observe

$$Y_n = u_f + \tfrac{1}{\sqrt{n}}\,\mathbb{W},$$

where $\mathbb{W}$ is a Gaussian white noise on $L^2(D)$. For $r \in (0,1)$ let

$$\mathcal{F}(\alpha, r) = \{h \in C^\alpha(D) : \ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r\},$$

where $\mathbf{1}$ is the constant function 1. For small but fixed $r \in (0,1)$, the question is: what is the minimax rate, as $n \to \infty$ and up to logarithmic factors, of

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\big[\|\hat{f}_n - f_0\|^2_{L^2(D)}\big] \ ?$$

4 Result

Theorem 4.1. *Fix $\alpha > 1$ and $d \geq 2$.*

(i) ***(Lower bound.)*** *For every $r \in (0,1)$ there are $c > 0$ and n_0 such that for $n \geq n_0$*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\hat{f} - f\|^2_{L^2(D)} \ \geq \ c\, n^{-2\alpha/(2\alpha+d+3)}.$$

(ii) ***(Upper bound.)*** *There is $r_0 = r_0(d,\alpha) > 0$ such that for every $r \in (0, r_0]$ an explicit estimator $\tilde{f}$ (Definition 8.2) satisfies, for $n \geq n_0$,*

$$\sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\tilde{f} - f\|^2_{L^2(D)} \ \leq \ C(\log n)^4\, n^{-2\alpha/(2\alpha+d+3)}.$$

If α is not an integer, the factor $(\log n)^4$ can be dropped.

The constants r_0, c, C, n_0 depend only on d, α, r and on fixed auxiliary profiles and kernels chosen below.

Corollary 4.2 (Answer)**.** *For every fixed $r \in (0, r_0]$, up to logarithmic factors,*

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_n - f_0\|^2_{L^2(D)} \ \asymp \ n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

Everything in Sections 7–8 is argued here; the only external inputs are the standard facts collected, with references, in Section 6.

5 Overview of the proof

Where $d+3$ comes from. Near $f = \mathbf{1}$ the solution is close to $u_1 := (|x|^2-1)/(2d)$, so $\nabla u_f \approx x/d$, a radial field with a single source at the origin. The linearised forward map $h \mapsto -L_f^{-1}\operatorname{div}(h\nabla u_f)$, with $L_f = \operatorname{div}(f\nabla\cdot)$ (Dirichlet), therefore smooths by order 2 for perturbations oscillating across the rays (in the $d-1$ angular directions), but only by order 1 along the rays, because $\operatorname{div}(h\,x/d)$ is a first-order operator along the rays. Heuristically, the symbol of the linearisation at $f = \mathbf{1}$ is $\asymp (1+|\xi_{\mathrm{rad}}|)/|\xi|^2$, and the exponent $d+3$ lies between the isotropic values $d+4$ and $d+2$. This heuristic only motivates the construction; it is not used in the proofs.

Lower bound. We perturb $\mathbf{1}$ by $M \asymp 2^{j(d-1)}$ disjoint bumps that oscillate at frequency 2^j in the angular directions only. Each has the form $2^{-j(\alpha+2)}b(|x|)\Delta\Phi_k$ with Φ_k homogeneous of degree 0; because of this form, the radial transport term $\operatorname{div}(h\,x/d)$ is again a Laplacian of a function of size $2^{-j(\alpha+2)}$. A duality argument then shows that the forward map shrinks each bump by 2^{-2j} in L^2. Balancing the Kullback–Leibler divergence with Assouad's lemma gives $2^j \asymp n^{1/(2\alpha+d+3)}$ and risk $2^{-2j\alpha}$.

Upper bound. We smooth the data at resolution 2^{-J}, $2^J \asymp n^{1/(2\alpha+d+3)}$, and estimate ∇u by $\hat V = \nabla K_J Y$. For r small, $\hat V$ is C^1-close to x/d, and the transport equation $\operatorname{div}(g\hat V) = 1$ has an explicit solution $T_{\hat V}^{-1}\mathbf{1}$, obtained by integrating along the backward flow of $\hat V$, which runs into a single sink. We linearise around the deterministic centring function $\bar f = T_{V_J}^{-1}\mathbf{1}$, where $V_J = \nabla K_J u$. This gives an exact decomposition: a bias $\bar f - f = O(J^2 2^{-J\alpha})$, a linear Gaussian term Z and a quadratic remainder. The variance of Z is the key point. Z integrates $\Delta\xi$ along rays, where ξ is the smoothed noise. Now $\Delta\xi$ has pointwise variance $n^{-1}2^{J(d+4)}$ and correlation length 2^{-J}, and integrating it along a ray of length $O(1)$ gains one factor 2^{-J}. Hence $\mathbb{E}\|Z\|_2^2 \lesssim n^{-1}2^{J(d+3)}$: the noise is smoothed by one order along the flow, and this produces $d+3$. The quadratic remainder is negligible because $\alpha > 1$.

6 Conventions and standard facts

C^s denotes the Hölder–Zygmund space $B^s_{\infty\infty}$; on D it consists of restrictions, with the quotient norm. Constants c, C may change from line to line. They never depend on j, J, n or on $f \in \mathcal{F}(\alpha, r)$. D° denotes the open unit ball. We write $u_1 := (|x|^2-1)/(2d)$, the solution for $f = \mathbf{1}$, so that $\nabla u_1 = x/d$. For $g \geq 1-r$ write $L_g = \operatorname{div}(g\nabla\cdot)$ with Dirichlet boundary conditions. Expectation is $\mathbb{E}$, and P_f is the law of Y_n. In the white-noise model, by the Cameron–Martin theorem (see e.g. [6]),

$$\mathrm{KL}(P_f, P_g) = \tfrac{n}{2}\|u_f - u_g\|_{L^2}^2.$$

We use the following known results without proof.

(E1) *Global H^2 regularity* [5, Theorem 8.12], combined with the energy estimate. Let $g \in C^{0,1}(D)$ with $g \geq 1-r$. The solution $w \in H_0^1(D)$ of $\operatorname{div}(g\nabla w) = F \in L^2$ satisfies $\|w\|_{H^2} \leq C\|F\|_2$, where C depends only on d, $1-r$ and an upper bound for $\|g\|_{C^{0,1}}$.

(E2) *$W^{2,p}$ theory*, $1 < p < \infty$ [5, Theorem 9.15, Lemma 9.17]. Let $g \in C^1(D)$ with $g \geq 1-r$. The Dirichlet problem $g\Delta w + \nabla g\cdot\nabla w = F \in L^p$, $w \in W^{2,p} \cap W_0^{1,p}$, has a unique solution, and $\|w\|_{W^{2,p}} \leq C\|F\|_p$. The constant is uniform over $\mathcal{F}(\alpha, r)$, which is bounded in C^α and hence precompact in C^1.

(GN) *Gagliardo–Nirenberg inequality on the ball* [4, 8]. For $p > d$ and $t = (1 + d/p)/2$, $\|\nabla w\|_\infty \le C\|w\|_{W^{2,p}}^t \|w\|_p^{1-t} + C\|w\|_p$.

(As) *Assouad's lemma* [1]; see also [12, Ch. 2]. Let $\{P_\theta : \theta \in \{0,1\}^M\}$ be probability measures with $\mathrm{TV}(P_\theta, P_{\theta'}) \le \tau < 1$ whenever θ, θ' differ in exactly one coordinate. Then for every estimator $\hat\theta$ with values in $\{0,1\}^M$, $\max_\theta \mathbb{E}_\theta \sum_k \mathbf{1}\{\hat\theta_k \ne \theta_k\} \ge \frac{M}{2}(1-\tau)$. We also use Pinsker's inequality $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$ [12, Ch. 2].

(C1) *Seeley-type extension* [9]; see also [10]. There is a linear operator E with $Eg = g$ on D that is bounded $C^s(D) \to C^s(\mathbb{R}^d)$ for $0 < s \le S_0$, where we fix $S_0 > \alpha + 2$. It has the form $Eg(t) = \chi(|t|) \sum_i a_i\, g(R_i t)$ for $|t| > 1$, where $R_i t = (1 - \lambda_i(|t| - 1))\, t/|t|$, finitely many $\lambda_i > 0$, and χ is a smooth cutoff vanishing for $|t| \ge 1 + \eta$, with $\lambda_i \eta < 1$.

(C2) *Kernel approximation* (see e.g. [6]). Let $k \in C_c^\infty(\mathbb{R}^d)$ with $\int k = 1$ and vanishing moments of orders $1, \dots, N_0$, where we fix an integer $N_0 \ge 2\alpha + 2$. Put $k_J = 2^{Jd} k(2^J \cdot)$. Then $\|D^m(k_J * g - g)\|_\infty \le C 2^{-J(s-m)} \|g\|_{C^s}$ for integers $0 \le m < s$ and $0 < s \le N_0$.

(LP) *Littlewood–Paley theory* [10, 6]. For a fixed Littlewood–Paley decomposition $g = \sum_{j \ge 0} \Delta_j g$ we have $\|g\|_{C^s} \asymp \sup_j 2^{js} \|\Delta_j g\|_\infty$ for $s > 0$. Bernstein's inequality gives $\|D^m \Delta_j g\|_\infty \le C 2^{jm} \|\Delta_j g\|_\infty$. Moreover $C^s \subset C^{1,\beta_0}$ for $s > 1$, with $\beta_0 = \min(s - 1, 1/2)$.

> **Reviewer 1**
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> Here and later: $C^{m,\alpha}$ never introduced.

(C3) *Taylor formula in Zygmund spaces; algebra property* (standard; see [10]). Let $\phi \in C^s$, $s > 0$, and $|h| \le 1$. Then $\Delta_h \phi(x) := \phi(x + h) - \phi(x) = P_\phi(x, h) + r_\phi(x, h)$, where $P_\phi(x, h) = \sum_{1 \le |\iota| < s} \partial^\iota \phi(x) h^\iota / \iota!$ and $|r_\phi(x, h)| \le C \|\phi\|_{C^s} |h|^s \Lambda(h)$ with $\Lambda(h) = 1 + \log(1/|h|)$; the logarithm is needed only when s is an integer. Also $C^s \cdot C^s \subset C^s$, with $\|\phi\psi\|_{C^s} \le C \|\phi\|_{C^s} \|\psi\|_{C^s}$.

(S) *Schauder estimate in Zygmund spaces* (standard). The Dirichlet Laplacian satisfies: $\Delta_D^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ is bounded for $\alpha - 1 > 0$. For non-integer α this is classical Schauder theory [5, Ch. 6].

> **Reviewer 1**
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> What about integer α?

(ODE) *C^k dependence of flows on initial data*, with the variational equations [7, Ch. V].

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> Details should be specified.

(G) *Gaussian facts* [6]. Let $y \mapsto \varphi_y \in L^2(D)$ be C^{m+1} on an open set, with locally Lipschitz derivatives. Then $\mathbb{W}(\varphi_y)$ has a C^m version with $\partial_y^\iota \mathbb{W}(\varphi_y) = \mathbb{W}(\partial_y^\iota \varphi_y)$ for $|\iota| \le m$. Bochner integrals in L^2 commute with $\mathbb{W}$. We also use Dudley's entropy bound [3] and the Borell–Tsirelson–Ibragimov–Sudakov (Borell–TIS) inequality [2, 11].

7 The lower bound

Throughout this section $\alpha > 1$, $d \ge 2$ and $r \in (0, 1)$ are fixed, and $j \ge j_0$ is large.

7.1 The hypercube

Directions and bumps. Choose $\omega_1, \dots, \omega_M \in S^{d-1}$ with pairwise geodesic distance $\geq 2 \cdot 2^{-j}$ and $M \geq c_d 2^{j(d-1)}$ (a maximal separated set). Fix $\varphi_0 \in C_c^\infty([0, 1/2))$ that is constant near 0 and not identically 0. Let $\Phi_k(\omega) = \varphi_0(2^j d_S(\omega, \omega_k))$, where d_S is the geodesic distance. The Φ_k are smooth, are rotations of one another, and have pairwise disjoint supports, which are caps of radius $2^{-j}/2$.

Extension and scaling. Extend each Φ_k to $\mathbb{R}^d \setminus \{0\}$ as a function homogeneous of degree 0. Then

$$\Delta \Phi_k(x) = |x|^{-2} (\Delta_S \Phi_k)(x/|x|),$$

where Δ_S is the Laplace–Beltrami operator. Put $\chi_k := 2^{-2j} \Delta_S \Phi_k$. In rescaled normal coordinates this is the flat Laplacian of a fixed nonzero compactly supported bump, up to $O(2^{-j})$ metric corrections. Hence $\|\partial^m \chi_k\|_\infty \leq C_m 2^{jm}$, and for $j \geq j_0$

> **Reviewer 1**
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> χ_k is multivariate and ∂^m for integer m is not well defined, please use multi-indices!

$$c 2^{-j(d-1)} \leq \|\chi_k\|^2_{L^2(S^{d-1})} \leq C 2^{-j(d-1)}.$$

The lower bound holds because the Laplacian of a nonzero compactly supported function is nonzero, and the value does not depend on k by rotation invariance.

The perturbations. Fix $b \in C_c^\infty((1/4, 3/4))$, not identically 0. Define

$$h_k(x) := \kappa 2^{-j(\alpha+2)} b(|x|)\, \Delta \Phi_k(x) = \kappa 2^{-j\alpha} b(|x|) |x|^{-2} \chi_k(x/|x|), \qquad f_\theta := \mathbf{1} + \sum_{k=1}^{M} \theta_k h_k, \quad \theta \in \{0,1\}^M.$$

Lemma 7.1. *There is $\kappa = \kappa(r) > 0$ such that $f_\theta \in \mathcal{F}(\alpha, r)$ for all θ and all j. The supports S_k of the h_k are pairwise disjoint, and*

$$\|h_k\|_2^2 \asymp 2^{-j(2\alpha+d-1)}, \qquad \|\Phi_k\|_{L^2(\{1/4<|x|<3/4\})} \asymp 2^{-j(d-1)/2}, \qquad \Big\| \sum_k \theta_k \Phi_k \Big\|_\infty \leq \|\varphi_0\|_\infty.$$

Proof. Let $G_\theta = \sum_k \theta_k \chi_k$. Each point of the sphere has a neighbourhood meeting at most one support, so $\|D^m G_\theta\|_\infty \leq C 2^{jm}$ and $\|G_\theta\|_\infty \leq C$. Take an integer $m > \alpha$. Then $|\Delta_h^m (2^{-j\alpha} G_\theta)| \leq C 2^{-j\alpha} \min((2^j|h|)^m, 1) \leq C|h|^\alpha$. Hence $\|2^{-j\alpha} G_\theta\|_{C^\alpha(S^{d-1})} \leq C$ uniformly in θ and j. Composing with $x \mapsto x/|x|$ and multiplying by the smooth function $b(|x|)|x|^{-2}$, supported in the annulus, keeps this bounded. Choosing κ small gives $\|f_\theta - \mathbf{1}\|_{C^\alpha(D)} \leq r$. Disjointness of the supports is clear. In polar coordinates, $\|h_k\|_2^2 = \kappa^2 2^{-2j\alpha} \int b(\rho)^2 \rho^{d-5} d\rho\, \|\chi_k\|^2_{L^2(S^{d-1})} \asymp 2^{-j(2\alpha+d-1)}$. The remaining two statements are immediate from the construction. □

By disjointness, $\|f_\theta - f_{\theta'}\|_2^2 = \sum_{k:\theta_k \neq \theta'_k} \|h_k\|_2^2$.

7.2 The key estimate

Proposition 7.2. *If $\theta, \theta' \in \{0,1\}^M$ differ only in coordinate k, then $\|u_{f_\theta} - u_{f_{\theta'}}\|_2 \leq C 2^{-2j} \|h_k\|_2$. Consequently $\|u_{f_\theta} - u_{f_{\theta'}}\|_2^2 \leq C 2^{-j(2\alpha+d+3)}$.*

Proof. Exact identity. Let $f = f_{\theta'}$, $g = f_\theta = f + h$ with $h = \pm h_k$, and $V = \nabla u_f$. Subtracting the two equations gives $\operatorname{div}(g\nabla(u_g - u_f)) = 1 - \operatorname{div}(g\nabla u_f) = -\operatorname{div}(hV)$. Hence

$$u_g - u_f = -L_g^{-1} \operatorname{div}(hV).$$

(i) Duality. L_g is self-adjoint on H_0^1, so $\|L_g^{-1}F\|_2 = \sup_{\|\phi\|_2 \le 1} \langle F, \psi\rangle$ with $\psi := L_g^{-1}\phi$. Since $g \ge 1 - r$ and g is Lipschitz (as $\alpha > 1$), (E1) gives $\|\psi\|_{H^2} \le C$. It therefore suffices to show

$$|\langle \operatorname{div}(h_k V), \psi\rangle| \le C 2^{-2j} \|h_k\|_2 \|\psi\|_{H^2} \qquad \text{for } \psi \in H^2 \cap H_0^1. \tag{1}$$

Split $V = x/d + \nabla v$ with $v := u_f - u_1$.

(ii) Radial part. For $G = b(|x|)\Delta\Phi_k$ we claim

$$\operatorname{div}(G\, x/d) = b_1(|x|)\Delta\Phi_k, \qquad b_1(\rho) := \frac{\rho b'(\rho) + (d-2)b(\rho)}{d} \in C_c^\infty((1/4, 3/4)). \tag{2}$$

Indeed, $\Psi := \Delta\Phi_k$ is homogeneous of degree -2, so Euler's relation gives $x \cdot \nabla G = \rho b' \Psi - 2b\Psi$.

> **Reviewer 1**
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> The last derivation is very quick and some intermediate steps should be given.

Since $\operatorname{div}(G\, x) = x \cdot \nabla G + dG$, dividing by d gives (2). The function $b_1\psi$ has compact support in the annulus, where Φ_k is smooth. Integrating by parts twice and using Lemma 7.1,

$$|\langle \operatorname{div}(h_k\, x/d), \psi\rangle| = \kappa 2^{-j(\alpha+2)} \Big| \int \Phi_k\, \Delta(b_1\psi) \Big| \le C 2^{-j(\alpha+2)} 2^{-j(d-1)/2} \|\psi\|_{H^2} \le C 2^{-2j} \|h_k\|_2 \|\psi\|_{H^2}.$$

(iii) Drift part. Since h_k has compact support, $|\langle \operatorname{div}(h_k \nabla v), \psi\rangle| = |\int h_k \nabla v \cdot \nabla\psi| \le \|h_k\|_2 \|\nabla v\|_\infty \|\psi\|_{H^1}$. It therefore suffices to show $\|\nabla v\|_\infty \le C 2^{-j(\alpha+1-d/p)}$ for every $p > d$, and then to take $p > d$ with $d/p \le \alpha - 1$.

Equation for v. We have $\operatorname{div}(f\nabla v) = 1 - \operatorname{div}(f\, x/d) = -\operatorname{div}((f-1)x/d) =: -q$. By (2) applied termwise, $q = \kappa 2^{-j(\alpha+2)} b_1(|x|)\Delta\Phi_{\theta'}$ with $\Phi_{\theta'} = \sum_l \theta'_l \Phi_l$.

$W^{2,p}$ bound. Since $q = \kappa 2^{-j\alpha} b_1(|x|)|x|^{-2} \sum_l \theta'_l \chi_l(x/|x|)$ and the χ_l are bounded with disjoint supports, $\|q\|_p \le C 2^{-j\alpha}$. Hence $\|v\|_{W^{2,p}} \le C 2^{-j\alpha}$ by (E2).

L^p bound. Take ϕ with $\|\phi\|_{p'} \le 1$ and let $\psi = L_f^{-1}\phi \in W^{2,p'} \cap W_0^{1,p'}$, so that $\|\psi\|_{W^{2,p'}} \le C$ by (E2). By Green's identity, and integrating by parts twice as in (ii),

$$\Big| \int v\phi \Big| = \Big| \int q\psi \Big| = \kappa 2^{-j(\alpha+2)} \Big| \int \Phi_{\theta'}\, \Delta(b_1\psi) \Big| \le C 2^{-j(\alpha+2)} \|\varphi_0\|_\infty \|\psi\|_{W^{2,p'}} \le C 2^{-j(\alpha+2)}.$$

So $\|v\|_p \le C 2^{-j(\alpha+2)}$.

Interpolation. By (GN), $\|\nabla v\|_\infty \le C 2^{-j\alpha t} 2^{-j(\alpha+2)(1-t)} + C 2^{-j(\alpha+2)} \le C 2^{-j(\alpha+1-d/p)}$, since $2(1-t) = 1 - d/p$.

Combining (ii) and (iii) proves (1), hence the first claim. The second follows from $\|h_k\|_2^2 \le C 2^{-j(2\alpha+d-1)}$. □

7.3 Conclusion via Assouad's lemma

Proof of Theorem 4.1(i). By Proposition 7.2, $\mathrm{KL}(P_{f_\theta}, P_{f_{\theta'}}) \le C n 2^{-j(2\alpha+d+3)}$ for neighbours θ, θ'. Choose $2^j \asymp n^{1/(2\alpha+d+3)}$, with a constant large enough that $\mathrm{KL} \le 1/8$ for all neighbours. Pinsker's inequality then gives $\mathrm{TV} \le 1/4$.

Given any estimator $\hat f$, let $\hat\theta_k$ minimise $\|(\hat f - 1 - th_k)\mathbf{1}_{S_k}\|_2$ over $t \in \{0,1\}$. On S_k we have $f_\theta = 1 + \theta_k h_k$. If $\hat\theta_k \neq \theta_k$, the triangle inequality gives $\|h_k\|_2 \le \|(\hat f - 1 - \hat\theta_k h_k)\mathbf{1}_{S_k}\|_2 + \|(\hat f - 1 - \theta_k h_k)\mathbf{1}_{S_k}\|_2 \le 2\|(\hat f - f_\theta)\mathbf{1}_{S_k}\|_2$. Since the S_k are disjoint,

$$\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \sum_k \|h_k\|_2^2\, \mathbf{1}\{\hat\theta_k \neq \theta_k\}.$$

By (As) with $\tau = 1/4$ and Lemma 7.1,

$$\sup_{f\in\mathcal F(\alpha,r)} \mathbb{E}_f\|\hat f - f\|_2^2 \ge \max_\theta \mathbb{E}_\theta\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \min_k \|h_k\|_2^2 \cdot \tfrac{M}{2}\cdot\tfrac34 \ge c2^{j(d-1)}2^{-j(2\alpha+d-1)} = c2^{-2j\alpha} \asymp n^{-2\alpha/(2\alpha+d+3)}.$$

The condition $\alpha > 1$ was used only in (iii), and to make $f \in C^1$ for (E1) and (E2). □

8 The upper bound

Throughout, $\alpha > 1$ and $d \ge 2$ are fixed, $f \in \mathcal F(\alpha, r)$, and $r \le r_0$, where r_0 is reduced finitely many times below.

8.1 A Schauder bound

Lemma 8.1. *There is $r_0 = r_0(d,\alpha)$ such that for $r \le r_0$ and all $f \in \mathcal F(\alpha, r)$,*

$$\|u_f - u_1\|_{C^{\alpha+1}(D)} \le Cr, \qquad \|u_f\|_{C^{\alpha+1}(D)} \le C. \tag{3}$$

Proof. Write $L_f = \Delta + \operatorname{div}((f-1)\nabla\cdot)$. By the algebra property (C3), $\|\operatorname{div}((f-1)\nabla v)\|_{C^{\alpha-1}} \le C_1\|f-1\|_{C^\alpha}\|v\|_{C^{\alpha+1}} \le C_1 r\|v\|_{C^{\alpha+1}}$. Take r_0 with $C_1 r_0\|\Delta_D^{-1}\| \le 1/2$. Writing $L_f v = F$ as $(I + \Delta_D^{-1}\operatorname{div}((f-1)\nabla\cdot))v = \Delta_D^{-1}F$ on the closed subspace of $C^{\alpha+1}(D)$ with zero trace, a Neumann series using (S) shows that $L_f^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ (Dirichlet data) has norm $\le 2\|\Delta_D^{-1}\|$. The solution so obtained is classical, hence equal to the weak solution. As in Proposition 7.2, $u_f - u_1 = -L_f^{-1}\operatorname{div}((f-1)x/d)$, and the right-hand side has $C^{\alpha-1}$ norm $\le Cr$. □

This is the first place where smallness of r enters.

8.2 Smoothing, noise and the estimator

Resolution and smoother. Let $J := \lfloor \log_2 n/(2\alpha + d + 3)\rfloor$. Then

$$n^{-1/2} \le 2^{-J(\alpha+(d+3)/2)}, \qquad n^{-1}2^{J(d+3)} \le 2^{-2J\alpha}, \qquad 2^{-2J\alpha} \le 2^{2\alpha} n^{-2\alpha/(2\alpha+d+3)}, \qquad J \le \log_2 n. \tag{4}$$

Put $K_J g := (k_J * Eg)|_D$, with k_J from (C2) and E from (C1). On data, $K_J Y(x) = \langle Y, k_J(x,\cdot)\rangle$, where $k_J(x,\cdot) := E^*[k_J(x-\cdot)] \in L^2(D)$; this is defined for all $x \in \mathbb{R}^d$. By the change of variables $z = R_i t$, E^* is the identity on D plus finitely many terms of the form $a_i(\chi\varphi)(R_i^{-1}z)|\det DR_i^{-1}(z)|$. These are supported on shells $\{1 - \lambda_i\eta < |z| < 1\}$ and have bounded Jacobians. This gives the following kernel bounds.

- **(K1)** $|\partial_x^\iota k_J(x,z)| \le C2^{J(d+|\iota|)}$ for $|\iota| \le 4$.
- **(K2)** $\partial_x^\iota k_J(x,z) = 0$ unless $|x - z| \le C_0 2^{-J}$. Indeed, for the reflected terms let $t = R_i^{-1}z$, so $|t| > 1$. If $x \in D$ and $|x - t| \le C2^{-J}$, then $|t| - 1 \le C2^{-J}$ and $|z - t| = (1+\lambda_i)(|t|-1) \le C2^{-J}$.
- **(K3)** $\|\partial_x^\iota k_J(x,\cdot)\|_2^2 \le C2^{J(d+2|\iota|)}$, and $x \mapsto \partial_x^\iota k_J(x,\cdot)$ is Lipschitz into L^2 with constant $C2^{J(d/2+|\iota|+1)}$, for $|\iota| \le 3$. This follows from (K1), (K2) and the fact that the support in z has volume $\le C2^{-Jd}$.

Notation. Let $U = Eu$, $F = Ef$, $V = \nabla u$ and $V^E = \nabla U$. By (C1) and (3), $\|F\|_{C^\alpha} + \|V^E\|_{C^\alpha} \le C$. Let

$$V_J := \nabla K_J u = (k_J * V^E)|_D, \qquad e := K_J u - u, \qquad \xi := n^{-1/2} K_J \mathbb{W}, \qquad \hat{V} := \nabla K_J Y = V_J + \nabla \xi.$$

V_J is the restriction of a C^∞ field on $\mathbb{R}^d$. By (G) and (K1)–(K3), ξ is a centred Gaussian field with C^3 sample paths on a neighbourhood of D (it is in fact smooth), and ξ does not depend on f.

Good event. Let

$$\mathcal{A} := \Big\{ \max_{|\iota|=m} \sup_D |\partial^\iota \xi| \le J n^{-1/2} 2^{J(d/2+m)},\ m = 1,2,3 \Big\}.$$

On $\mathcal{A}$, by (4) and after adjusting constants for operator norms,

$$\|\nabla \xi\|_\infty \le C J 2^{-J(\alpha+1/2)}, \qquad \|D^2 \xi\|_\infty \le C J 2^{-J(\alpha-1/2)}, \qquad \|D^3 \xi\|_\infty \le C J 2^{-J(\alpha-3/2)}. \tag{5}$$

Definition 8.2 (Estimator)**.** Let $\varepsilon_0 := 1/(8d)$, and define the observable event $\mathcal{B} := \{\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon_0\}$. On $\mathcal{B}$ set $\hat{f} := T_{\hat{V}}^{-1}\mathbf{1}$ (Definition 8.6); off $\mathcal{B}$ set $\hat{f} := \mathbf{1}$. Let $\tilde{f} := \min(\max(\hat{f}, 1-r), 1+r)$.

Since $\|f - 1\|_\infty \le r$, clipping can only decrease $|\tilde{f} - f|$ pointwise. The estimator uses only Y, n, α, r and the fixed kernels. The event $\mathcal{A}$ itself is not observable: ξ does not depend on f, but $\xi = K_J Y - K_J u$ cannot be computed from the data because u is unknown. This is why the estimator is defined on $\mathcal{B}$; Lemma 8.4 shows $\mathcal{A} \subset \mathcal{B}$.

8.3 The smoothed fields

Lemma 8.3. *Let $g \in C^s(\mathbb{R}^d)$, $s > 0$, and let $m \ge 1$ be an integer. Then $\|D^m(k_J * g)\|_\infty$ is at most $C 2^{J(m-s)} \|g\|_{C^s}$ if $m > s$, at most $C(J+1)\|g\|_{C^s}$ if $m = s$, and at most $C\|g\|_{C^s}$ if $m < s$.*

Proof. By (LP), $D^m(k_J * g) = \sum_j (D^m k_J) * \Delta_j g$, and the series converges uniformly because each term is $\le \|D^m k_J\|_1 \|\Delta_j g\|_\infty \le C 2^{Jm} 2^{-js} \|g\|_{C^s}$. For $j \le J$, write the term as $k_J * D^m \Delta_j g$; by Bernstein it is $\le C 2^{j(m-s)} \|g\|_{C^s}$. For $j > J$, the first bound gives $C 2^{Jm-js} \|g\|_{C^s}$, and summing over $j > J$ gives $C 2^{J(m-s)} \|g\|_{C^s}$. Summing the bound $C 2^{j(m-s)}$ over $j \le J$ gives the three cases. □

Lemma 8.4. *Let $\varepsilon \le \varepsilon_0$. There are r_0 and J_0 such that for $r \le r_0$ and $J \ge J_0$:*

(1) $\|V_J - x/d\|_{C^1(D)} \le \varepsilon/2$;

(2) on $\mathcal{A}$, $\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon$; in particular (with $\varepsilon = \varepsilon_0$) $\mathcal{A} \subset \mathcal{B}$;

(3) $\|D^m V_J\|_\infty \le C J 2^{J(m-\alpha)_+}$ for $m = 2, 3$, with no factor J unless $\alpha = m$.

Proof. Item (3) is Lemma 8.3 with $g = V^E$ and $s = \alpha$. For (1), on D we have $V_J - x/d = (k_J * V^E - V^E) + \nabla(u - u_1)$. By (LP), $D^i V^E$ is β_0-Hölder for $i = 0, 1$, with $\beta_0 = \min(\alpha - 1, 1/2)$. Hence $|D^i(k_J * V^E - V^E)| \le C \int |k_J(y)||y|^{\beta_0} dy \le C 2^{-J\beta_0}$. By (3), $\|\nabla(u - u_1)\|_{C^1} \le Cr$. Take r_0 and J_0 such that $C(2^{-J_0\beta_0} + r_0) \le \varepsilon/2$. For (2), on $\mathcal{A}$, (5) gives $\|\nabla\xi\|_{C^1} \le C J 2^{-J(\alpha-1/2)} \le \varepsilon/2$ for $J \ge J_0$. □

8.4 Geometry of the flow and the transport inverse

Setting. Let X be the restriction to D of a C^k field ($k \ge 1$) on a neighbourhood of D, with

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> Double use of symbol 'D', comment or rename.

$$\|X - x/d\|_{C^1(D)} := \max\big(\sup_D |X - x/d|,\ \sup_D \|DX - I/d\|\big) \le \varepsilon \le 1/(8d).$$

Put $\nu := 1/d - \varepsilon \ge 7/(8d)$. Let Φ_t be the flow of X, let $\gamma(s) = \gamma_x(s) = \Phi_{-s}x$ be the backward ray from x, and let

$$w(s,x) := \exp\Big(-\int_0^s \operatorname{div} X(\Phi_{-\sigma}x)\,d\sigma\Big).$$

Lemma 8.5. *In this setting the following hold.*

(1) X has exactly one zero $x_X \in D$, and $|x_X| \le d\varepsilon$.

(2) With $\rho(y) = |y - x_X|$: $(y - x_X)\cdot X(y) \ge (1/d - \varepsilon)\rho^2$ and $|X(y)| \le (1/d + \varepsilon)\rho$ on D.

(3) For $x \in D$, $\Phi_{-s}x$ exists for all $s \ge 0$ and stays in D. It lies in D° for $s > 0$, and D° is invariant.

(4) $\varrho(s) := \rho(\gamma(s))$ satisfies $\varrho(0)e^{-(1/d+\varepsilon)s} \le \varrho(s) \le \varrho(0)e^{-(1/d-\varepsilon)s}$. If $\varrho(0) > 0$, then $-(1/d + \varepsilon)\varrho \le \varrho' \le -\nu\varrho$.

(5) The propagator $P(s,\sigma)$ of $y' = -DX(\gamma)y$ satisfies $\|P(s,\sigma)\| \le e^{-\nu(s-\sigma)}$. In particular $\|D\Phi_{-s}\| \le e^{-\nu s}$.

(6) $1 - d\varepsilon \le \operatorname{div} X \le 1 + d\varepsilon$, so $e^{-(1+d\varepsilon)s} \le w \le e^{-(1-d\varepsilon)s} \le e^{-7s/8}$.

Proof.

> **Reviewer 1**
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> This proof is *very* dense and parts require some intermediate steps or references.

(1) The map $x \mapsto x - dX(x) = -d(X - x/d)$ sends D into $B(0, d\varepsilon) \subset D$ and is a $d\varepsilon$-contraction on the convex set D. Its unique fixed point is the unique zero of X. (2) $X(y) = Q(y)(y - x_X)$ with $Q = \int_0^1 DX(x_X + t(y - x_X))\,dt$ and $\|Q - I/d\| \le \varepsilon$; the segment lies in D by convexity. (3) For $|z| = 1$, $z \cdot X(z) \ge 1/d - \varepsilon > 0$, so $\frac{d}{ds}|\gamma|^2 = -2\gamma \cdot X(\gamma) < 0$ whenever $|\gamma| = 1$. A first-exit-time argument gives invariance of D and of D°, and shows that boundary points enter D° immediately. Global existence holds because D is compact. (4) $(\varrho^2)' = -2(\gamma - x_X)\cdot X(\gamma)$; apply (2) and Gronwall's inequality. (5) $v \cdot DX\,v \ge \nu|v|^2$ on D, so $(|y|^2)' \le -2\nu|y|^2$. Also $D\Phi_{-s} = P(s,0)$. (6) $\operatorname{div} X = 1 + \operatorname{tr}(DX - I/d)$ and $|\operatorname{tr} A| \le d\|A\|$. □

Definition 8.6. For $g \in C^1(D)$ put $T_X g := \operatorname{div}(gX)$. For $H \in C(D)$ put

$$T_X^{-1}H(x) := \int_0^\infty H(\Phi_{-s}x)\,w(s,x)\,ds.$$

By Lemma 8.5(6) the integral converges absolutely and uniformly, and $T_X^{-1}H \in C(D)$.

Since $w(s, \Phi_{-t}x) = w(s+t,x)/w(t,x)$, we have the semigroup identity

$$w(t,x)\,T_X^{-1}H(\Phi_{-t}x) = \int_t^\infty H(\Phi_{-s}x)\,w(s,x)\,ds. \tag{6}$$

Lemma 8.7. *In the setting of Lemma 8.5:*

(a) (Uniqueness.) *If* $g \in C^1(D)$ *and* $\operatorname{div}(gX) = H$ *on* D° *with* $H \in C(D)$*, then* $g = T_X^{-1}H$*.*

(a′) (Right inverse.) *If* $H \in C(D)$ *and* $g := T_X^{-1}H \in C^1(D)$*, then* $\operatorname{div}(gX) = H$ *on* D*.*

(b) $\|T_X^{-1}H\|_\infty \le \frac{8}{7}\|H\|_\infty \le 2\|H\|_\infty$*.*

(c) (Boundary layer.) *If* $H = 0$ *on* $\{|x| \le 1-\delta\}$*, then* $\|T_X^{-1}H\|_\infty \le 6d\delta\|H\|_\infty$*.*

Proof. (a) Fix $x \in D$. For $s > 0$, $\gamma(s) \in D^\circ$, and $\psi(s) = g(\gamma(s))w(s,x)$ satisfies $\psi' = -(X\cdot\nabla g + g\operatorname{div}X)(\gamma)\,w = -H(\gamma)\,w$. Integrate over $[\delta, S]$, then let $\delta \to 0$ (by continuity) and $S \to \infty$ (using $|\psi(S)| \le \|g\|_\infty e^{-7S/8}$). This gives $g(x) = \int_0^\infty H(\gamma(s))w(s,x)\,ds$.

(a′) Fix $x \in D^\circ$ and differentiate (6) at $t = 0^+$. The right side gives $-H(x)$. Since $w(0,x) = 1$ and $\partial_t w(0,x) = -\operatorname{div}X(x)$, the left side gives $-\nabla g(x)\cdot X(x) - g(x)\operatorname{div}X(x)$. Extend to D by continuity.

(b) Use $w \le e^{-7s/8}$.

(c) The case $\delta > 1/2$ follows from (b), since then $6d\delta > 2$. Let $\delta \le 1/2$ and $\phi = |\gamma|^2$. Where $|\gamma| \ge 1/2$, $\gamma\cdot X(\gamma) \ge |\gamma|(|\gamma|/d - \varepsilon) \ge 3/(16d)$, so $\phi' \le -3/(8d)$. Hence ϕ drops below $(1-\delta)^2$ within time $\le (1-(1-\delta)^2)\cdot 8d/3 \le 6d\delta$, and by the same sign argument it never returns. Since $H(\gamma(s)) = 0$ afterwards and $w \le 1$, the claim follows. □

Lemma 8.8 (Flow derivatives, uniform in s)**.** *Let* $M = D\Phi_{-s}$*,* $N = D^2\Phi_{-s}$ *and* $\ell = \log w$*, all differentiated in* x*.*

(1) If $X \in C^2$*:* $\|N(s)\| \le \|D^2X\|_\infty e^{-\nu s}/\nu$ *and* $|D_x\ell| \le \|D\operatorname{div}X\|_\infty/\nu$*.*

(2) If $X \in C^3$*:* $\|D_x^2\ell\| \le \|D^2\operatorname{div}X\|_\infty/(2\nu) + \|D\operatorname{div}X\|_\infty\|D^2X\|_\infty/\nu^2$*.*

Proof. N solves $N' = -DX(\gamma)N - D^2X(\gamma)[M\cdot, M\cdot]$ with $N(0) = 0$. By Duhamel's formula and Lemma 8.5(5), $\|N(s)\| \le \int_0^s e^{-\nu(s-\sigma)}\|D^2X\|_\infty e^{-2\nu\sigma}d\sigma \le \|D^2X\|_\infty e^{-\nu s}/\nu$. Next, $D_x\ell[v] = -\int_0^s D\operatorname{div}X(\gamma)[Mv]\,d\sigma$, and $\int_0^\infty e^{-\nu\sigma}d\sigma = 1/\nu$. Finally, $D_x^2\ell[v,v'] = -\int_0^s \big(D^2\operatorname{div}X(\gamma)[Mv, Mv'] + D\operatorname{div}X(\gamma)[N[v,v']]\big)d\sigma$; bound this with the two previous estimates. □

Differentiation under the integral. Let the integrand be $H(\Phi_{-s}x)w(s,x)$ or $w(s,x)$. By (ODE), and since the flow is defined on a neighbourhood of D, its x-derivatives up to the relevant order are jointly continuous on $[0,\infty)\times D$. By Lemma 8.8 and the identities $D_xw = w\,D_x\ell$ and $D_x^2w = w(D_x^2\ell + D_x\ell\otimes D_x\ell)$, they are dominated by $Ke^{-7s/8}$. Truncating the integral at S, differentiating, and passing to the uniform limit gives the following.

Lemma 8.9. *All constants below depend only on* d*, uniformly over* $\|X - x/d\|_{C^1} \le \varepsilon \le 1/(8d)$*.*

(d) *If* $X \in C^2$ *and* $H \in C^1(D)$*, then* $T_X^{-1}H \in C^1(D)$ *and* $\|\nabla T_X^{-1}H\|_\infty \le C_d(\|\nabla H\|_\infty + \|H\|_\infty\|D\operatorname{div}X\|_\infty)$*.*

(e) *If* $X \in C^3$*, then* $T_X^{-1}\mathbf{1} \in C^2(D)$*,* $\|\nabla T_X^{-1}\mathbf{1}\|_\infty \le C_d\|D\operatorname{div}X\|_\infty$*, and* $\|D^2T_X^{-1}\mathbf{1}\|_\infty \le C_d\big(\|D^2\operatorname{div}X\|_\infty + \|D\operatorname{div}X\|_\infty^2 + \|D\operatorname{div}X\|_\infty\|D^2X\|_\infty\big)$*.*

8.5 The centring function and an exact decomposition

Fix $\varepsilon = \varepsilon_0$ in Lemma 8.4, so that Lemmas 8.5–8.9 apply to $X = V_J$ (with $\varepsilon_0/2$), and on $\mathcal{A}$ also to $X = \hat{V}$. Let

$$\bar{f} := T_{V_J}^{-1}\mathbf{1}.$$

Using $\|D^m \operatorname{div} V_J\|_\infty \le d\|D^{m+1}V_J\|_\infty$, Lemma 8.4(3) and Lemma 8.9(e),

$$\|\bar f\|_\infty \le 2, \qquad \|\nabla \bar f\|_\infty \le CJ2^{J(2-\alpha)_+}, \qquad \|D^2 \bar f\|_\infty \le CJ^2 2^{J(3-\alpha)_+}. \tag{7}$$

The last bound uses $2(2-\alpha)_+ \le (3-\alpha)_+$ for $\alpha \ge 1$. Moreover $\bar f \in C^2(D)$, and $\operatorname{div}(\bar f V_J) = 1$ by Lemma 8.7(a′).

Noise term. Let $G_\xi := \operatorname{div}(\bar f \nabla \xi) = \bar f \Delta\xi + \nabla \bar f \cdot \nabla \xi \in C^1(D)$ and $Z := T_{V_J}^{-1} G_\xi$. Since V_J is deterministic, Z is defined for every realisation. By Lemma 8.9(d), $Z \in C^1(D)$, and $T_{V_J} Z = G_\xi$ by Lemma 8.7(a′).

Proposition 8.10. *On $\mathcal{A}$ (so that $\hat V$ satisfies the hypotheses of Lemma 8.5 and $\hat f = T_{\hat V}^{-1}\mathbf{1}$),*

$$\hat f - f = (\bar f - f) - Z + T_{\hat V}^{-1} \operatorname{div}(Z \nabla \xi). \tag{8}$$

Proof. $\hat V$ is C^3 on a neighbourhood of D, so $\hat f \in C^2$ by Lemma 8.9(e), and $T_{\hat V}\hat f = 1$ by Lemma 8.7(a′). Hence $T_{\hat V}(\hat f - \bar f) = 1 - \operatorname{div}(\bar f V_J) - \operatorname{div}(\bar f \nabla \xi) = -G_\xi$. Also $T_{\hat V} Z = T_{V_J} Z + \operatorname{div}(Z\nabla\xi) = G_\xi + \operatorname{div}(Z\nabla\xi)$. Therefore $T_{\hat V}(\hat f - \bar f + Z) = \operatorname{div}(Z\nabla\xi)$, which is continuous. Since $\hat f - \bar f + Z \in C^1$, Lemma 8.7(a) with $X = \hat V$ gives $\hat f - \bar f + Z = T_{\hat V}^{-1} \operatorname{div}(Z\nabla\xi)$, which is (8). □

8.6 The bias: $\|\bar f - f\|_\infty \le CJ^2 2^{-J\alpha}$

Proposition 8.11. *Uniformly over $f \in \mathcal{F}(\alpha, r)$, $\|\bar f - f\|_\infty \le CJ^2 2^{-J\alpha}$, with no factor J^2 when $\alpha \notin \mathbb{N}$.*

Proof. Reduction. $f \in C^1$ (as $\alpha > 1$), $u \in C^{\alpha+1} \subset C^2$ by (3), $\operatorname{div}(fV) = 1$, and $V_J - V = \nabla e$ on D. So $T_{V_J}(\bar f - f) = 1 - \operatorname{div}(fV_J) = -\operatorname{div}(f\nabla e)$, and by Lemma 8.7(a), $\bar f - f = -T_{V_J}^{-1}\operatorname{div}(f\nabla e)$.

Commutator. Let $g := F - k_J * F$ and $R(x) := -\int k_J(-h)\Delta_h F(x)\Delta_h V^E(x)\,dh$. Then

$$F\,(k_J * V^E) = k_J * (FV^E) + gV^E + R.$$

Indeed, $F(x)(k_J * V^E)(x) - k_J * (FV^E)(x) = \int k_J(y)[F(x) - F(x-y)]V^E(x-y)\,dy$. Insert $F(x) - F(x-y) = -\Delta_{-y}F(x)$ and $V^E(x-y) = V^E(x) + \Delta_{-y}V^E(x)$, and use $\int k_J = 1$; the first part gives gV^E and the second, after $h = -y$, gives R. Let $\Gamma := \operatorname{div}(FV^E) - 1$, which lies in $C^{\alpha-1}(\mathbb{R}^d)$ by the algebra property and vanishes on D. Since $\operatorname{div}(k_J * (FV^E)) = k_J * \Gamma + 1$, on D

$$\operatorname{div}(f\nabla e) = \operatorname{div}(fV_J) - 1 = k_J * \Gamma + \operatorname{div}(gV) + \operatorname{div} R.$$

We treat the three terms in turn, using linearity of $T_{V_J}^{-1}$.

(i) The term $\operatorname{div}(gV)$. Since $V = V_J - \nabla e$, we have $\operatorname{div}(gV) = T_{V_J} g - \operatorname{div}(g\nabla e)$. As $g \in C^1$, Lemma 8.7(a) gives $T_{V_J}^{-1}\operatorname{div}(gV) = g - T_{V_J}^{-1}\operatorname{div}(g\nabla e)$. By (C2) (applied to $F \in C^\alpha$ and $U \in C^{\alpha+1}$): $\|g\|_\infty \le C2^{-J\alpha}$, $\|\nabla g\|_\infty \le C2^{-J(\alpha-1)}$, $\|\nabla e\|_\infty \le C2^{-J\alpha}$ and $\|D^2 e\|_\infty \le C2^{-J(\alpha-1)}$. By Lemma 8.7(b), the contribution is $\le C2^{-J\alpha} + C2^{-J(2\alpha-1)}$.

(ii) The term $\operatorname{div} R$. We have $\partial R = -\int k_J(-h)\big[\Delta_h \partial F\,\Delta_h V^E + \Delta_h F\,\Delta_h \partial V^E\big]dh$. Each product has the form $\Delta_h\phi_1\,\Delta_h\phi_2$ with $\phi_i \in C^{s_i}$ and $(s_1, s_2) = (\alpha-1, \alpha)$ or $(\alpha, \alpha-1)$. By (C3), and using $|P_{\phi_i}| \le C|h|$ for $|h| \le 1$, the product equals a polynomial in h (with coefficients depending on x) whose monomials have degrees in $[2, s_1+s_2) = [2, 2\alpha-1)$. The remainder is bounded by $C|h|^{\min(1+s_1,\,1+s_2,\,s_1+s_2)}\Lambda(h)^2 = C|h|^\alpha\Lambda(h)^2$. When $s_i < 1$, P_{ϕ_i} is empty. The polynomial terms integrate to 0 against $k_J(-h)$, because $N_0 \ge 2\alpha + 2$. Since k_J is supported in $|h| \le C2^{-J}$, the

substitution $h = 2^{-J}y$ gives $\int |k_J(h)||h|^\alpha \Lambda(h)^2 dh \le CJ^2 2^{-J\alpha}$. So $\|\operatorname{div} R\|_\infty \le CJ^2 2^{-J\alpha}$, with no J^2 for non-integer α (then no logarithm appears in (C3)). By Lemma 8.7(b), the contribution is $\le CJ^2 2^{-J\alpha}$.

(iii) The term $k_J * \Gamma$. On D, $k_J * \Gamma = k_J * \Gamma - \Gamma$, so by (C2) with $s = \alpha - 1$ and $m = 0$ it is $O(2^{-J(\alpha-1)})$. By the support of k_J, it vanishes at points of D at distance $> C2^{-J}$ from ∂D. By Lemma 8.7(c) with $\delta = C2^{-J}$, the contribution is $\le C2^{-J}2^{-J(\alpha-1)} = C2^{-J\alpha}$.

Adding (i)–(iii) gives the claim. □

8.7 The variance: $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$

Proposition 8.12. $\mathbb{E}\|Z\|_{L^2(D)}^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$.

Proof. Representation. Let $\mathcal{D}_y := \bar f(y)\Delta_y + \nabla \bar f(y)\cdot\nabla_y$. By (G), $G_\xi(y) = n^{-1/2}\mathbb{W}(\mathcal{D}_y k_J(y,\cdot))$, and since Bochner integrals commute with $\mathbb{W}$,

$$Z(x) = n^{-1/2}\mathbb{W}(h_x), \qquad h_x = \int_0^\infty w(s,x)\,\mathcal{D}_y k_J(y,\cdot)\big|_{y=\gamma_x(s)}\,ds,$$

where γ_x is the backward ray of V_J. Hence $\mathbb{E}Z(x)^2 = n^{-1}\|h_x\|_2^2$.

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> Define 'backward ray' mathematically, maybe 'characteristics' is the more appropriate term.

Kernel bound. Since $(2-\alpha)_+ < 1$, (7) gives $\|\nabla \bar f\|_\infty \le CJ2^{J(2-\alpha)_+} \le C2^J$. With (K1) and (K2), $|\mathcal{D}_y k_J(y,z)| \le C2^{J(d+2)}\mathbf{1}\{|y-z| \le \delta_J\}$, where $\delta_J := C_0 2^{-J}$. Let $\mu_x := \int_0^\infty w(s,x)\,\delta_{\gamma_x(s)}\,ds$, a measure on D of mass $\le 8/7$. Then

$$|h_x(z)| \le C2^{J(d+2)}\mu_x(B(z,\delta_J)), \qquad \|h_x\|_2^2 \le C2^{2J(d+2)}\int \mu_x(B(z,\delta_J))^2 dz \le C2^{2J(d+2)}\delta_J^d \int \mu_x(B(y,2\delta_J))\,\mu_x(dy).$$

The last step is Fubini: $\int \mu(B(z,\delta))^2 dz = \iint |B(y,\delta)\cap B(y',\delta)|\,\mu(dy)\mu(dy')$, and the intersection is empty unless $|y-y'| \le 2\delta$.

Ray measure. Let x_J be the zero of V_J, $\rho = |\cdot - x_J|$, and $\rho_x = \rho(x) > 0$. Let $\varrho(s) = \rho(\gamma_x(s))$. By Lemma 8.5(4),(6), $w(s,x) \le (\varrho(s)/\rho_x)^\beta$ with $\beta = d(1-d\varepsilon)/(1+d\varepsilon) \ge 7d/9 > 1$. Moreover ϱ decreases strictly from ρ_x to 0, with $|\varrho'| \ge \nu\varrho$. Changing variables $t = \varrho(s)$, for every $\varphi \ge 0$,

$$\int \varphi\,d\mu_x \le \frac{1}{\nu}\int_0^{\rho_x} \varphi(\gamma_x \text{ at radius } t)\,(t/\rho_x)^\beta\,\frac{dt}{t}. \tag{9}$$

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> What do you mean exactly by 'γ_x at radius t', why is it unique?

Small-ball mass. If $|\gamma_x(s) - y| \le 2\delta_J$, then $|\varrho(s) - \rho(y)| \le 2\delta_J$. Hence, by (9), $m(y) := \mu_x(B(y,2\delta_J))$ satisfies

$$m(y) \le C\delta_J \rho(y)^{\beta-1}\rho_x^{-\beta} \text{ if } \rho(y) \ge 4\delta_J, \qquad m(y) \le C\min(1,\delta_J/\rho_x)^\beta \text{ if } \rho(y) < 4\delta_J.$$

(In the first case $t \in [\rho(y)/2, 3\rho(y)/2]$ ranges over an interval of length $4\delta_J$; in the second $t \le \min(6\delta_J, \rho_x)$.)

Integration. Apply (9) once more, with $\varphi = m$. The part with $t \ge 4\delta_J$ is $\le C\delta_J\rho_x^{-2\beta}\int_0^{\rho_x} t^{2\beta-2}dt \le C\delta_J/\rho_x$, since $2\beta - 1 > 0$. The part with $t < 4\delta_J$ is $\le C\min(1,\delta_J/\rho_x)^{2\beta} \le C\min(1,\delta_J/\rho_x)$. Trivially, $\int m\,d\mu_x \le \mu_x(D)^2 \le 4$. Hence $\int m\,d\mu_x \le C\min(1,\delta_J/\rho_x)$, and

$$\|h_x\|_2^2 \le C2^{J(d+4)}\min(1, 2^{-J}/\rho_x) \le C2^{J(d+3)}|x - x_J|^{-1}.$$

Since $\int_D |x-x_J|^{-1}dx \le C$ for $d \ge 2$, Tonelli's theorem and (4) give $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$. □

8.8 The remainder and the bad event

Lemma 8.13. *On $\mathcal{A}$, $\|T_{\hat V}^{-1}\operatorname{div}(Z\nabla\xi)\|_\infty \le CJ^4 2^{-J(2\alpha-1)}$. This is $o(2^{-J\alpha})$ because $\alpha > 1$.*

Proof. We use (5), (7), $(2-\alpha)_+ < 1$ and $(3-\alpha)_+ \le 2$. First,

$$\|G_\xi\|_\infty \le \|\bar f\|_\infty\|\Delta\xi\|_\infty + \|\nabla\bar f\|_\infty\|\nabla\xi\|_\infty \le CJ2^{-J(\alpha-1/2)} + CJ^2 2^{J(2-\alpha)_+ - J(\alpha+1/2)} \le CJ^2 2^{-J(\alpha-1/2)}.$$

Next, $\|\nabla G_\xi\|_\infty \le C(\|D^3\xi\|_\infty + \|\nabla\bar f\|_\infty\|D^2\xi\|_\infty + \|D^2\bar f\|_\infty\|\nabla\xi\|_\infty) \le CJ^3 2^{-J(\alpha-3/2)}$. Here each cross term has exponent $\le -(\alpha-3/2)$ because $\alpha \ge 1$.

By Lemma 8.7(b) and Lemma 8.9(d), with $\|D\operatorname{div} V_J\|_\infty \le d\|D^2 V_J\|_\infty \le CJ2^{J(2-\alpha)_+}$ from Lemma 8.4(3), we get $\|Z\|_\infty \le CJ^2 2^{-J(\alpha-1/2)}$ and $\|\nabla Z\|_\infty \le C(\|\nabla G_\xi\|_\infty + \|G_\xi\|_\infty J2^{J(2-\alpha)_+}) \le CJ^3 2^{-J(\alpha-3/2)}$. Hence

$$\|\operatorname{div}(Z\nabla\xi)\|_\infty \le \|\nabla Z\|_\infty\|\nabla\xi\|_\infty + \|Z\|_\infty\|\Delta\xi\|_\infty \le CJ^4 2^{-J(2\alpha-1)} + CJ^3 2^{-J(2\alpha-1)} \le CJ^4 2^{-J(2\alpha-1)}.$$

Lemma 8.7(b) for $X = \hat V$ concludes the proof. □

Lemma 8.14. *$P(\mathcal{A}^c) \le Ce^{-cJ^2}$, uniformly in f.*

Proof. Each component $\partial^\iota\xi(x) = n^{-1/2}\mathbb{W}(\partial_x^\iota k_J(x,\cdot))$, $|\iota| = m$, is centred Gaussian. By (K3) its standard deviation is $\le C\bar\sigma_m$, where $\bar\sigma_m := n^{-1/2}2^{J(d/2+m)}$, and its canonical metric is $\le C\bar\sigma_m 2^J|x-x'|$. The covering numbers of D in this metric are therefore $\le (C\bar\sigma_m 2^J/\eta)^d$, and Dudley's bound gives $\mathbb{E}\sup_D|\partial^\iota\xi| \le C\bar\sigma_m\sqrt{J} \le J\bar\sigma_m/2$ for large J. The Borell–TIS inequality at deviation $J\bar\sigma_m/2$ gives $P(\sup_D|\partial^\iota\xi| > J\bar\sigma_m) \le 2e^{-cJ^2}$. A union bound over the finitely many ι with $1 \le |\iota| \le 3$ concludes the proof. □

8.9 Assembly

Proof of Theorem 4.1(ii). Let $r \le r_0$ (the minimum of the values required in Lemmas 8.1 and 8.4) and $J \ge J_0$. On $\mathcal{A}$ we have $\mathcal{A} \subset \mathcal{B}$, so (8) applies. Clipping only decreases $|\tilde f - f|$, so Proposition 8.11 and Lemma 8.13 give

$$\|\tilde f - f\|_2^2 \le 3|D|\big(CJ^2 2^{-J\alpha}\big)^2 + 3\|Z\|_2^2 + 3|D|\big(CJ^4 2^{-J(2\alpha-1)}\big)^2.$$

Off $\mathcal{A}$, $|\tilde f - f| \le 2r$ pointwise, so $\|\tilde f - f\|_2^2 \le 4r^2|D|$. Taking expectations, using Proposition 8.12, Lemma 8.14 and (4),

$$\mathbb{E}_f\|\tilde f - f\|_2^2 \le CJ^4 2^{-2J\alpha} + C2^{-2J\alpha} + CJ^8 2^{-2J(2\alpha-1)} + Ce^{-cJ^2} \le CJ^4 2^{-2J\alpha} \le C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)},$$

uniformly over $\mathcal{F}(\alpha, r)$. The finitely many n with $J < J_0$ are covered by the trivial bound $4r^2|D|$. For non-integer α, Lemma 8.3 and (C3) produce no logarithm. Then the bias is $\le C2^{-J\alpha}$, and the remaining powers of J multiply terms of strictly smaller order. So the bound is $Cn^{-2\alpha/(2\alpha+d+3)}$. □

Theorem 4.1 follows from Sections 7 and 8.

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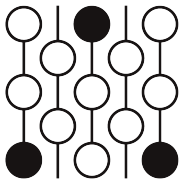
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Review of Problem 2 – Siege

First Proof Third Batch
Reviewer 2

October 8–9, 2026



1 Review

Summary. The manuscript combines the inversion formula for the Darcy problem with a plug-in kernel estimator for the solution field u_f to establish an upper bound on the estimation risk. This is accomplished without a pre-estimation step. The heuristic explanation of the rate in Section 5 is on point. On a technical level, the analysis combines properties of kernel estimators with stability properties of the inverse Darcy operator. The lower bound is established using the classical Assouad's lemma with perturbations of the diffusivity. The upper and lower bounds match up to logarithmic factors.
Novelty. The general approach to establishing the upper bounds relies on the inversion formula for the Darcy problem, as in the human solution. The use of this formula is itself novel in the field. The main difference from the human solution lies in the direct treatment of the L^2 risk, rather than the pointwise risk, and the use of a different error decomposition. There are also minor differences in the construction of the estimator and in the proof of the lower bounds, where the alternatives are constructed on the level of the conductivity f instead of u_f.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5
Justification: The overall estimation estimation and proof strategy are sound and use the correct technical tools. A full verification of the proofs is difficult due to overly complicated notation and structure.

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort

- -1: very poor exposition

Score: 0
Justification: The exposition is readable with some serious effort, mainly due to unclear structure with key connected statements split across several Lemmas. In multiple places definitions are used that are only introduced at a later point in time, for instance in the definition of the estimator. While the general strategy is explained in Section 5, key proof ideas are not clearly motivated or articulated. In some places (for example in 8.4, Def 8.6) the notation is unnecessarily complicated, or arguments are very brief (e.g. Proofs of Lemma 8.5 (6), Lemma 8.7 (b)).

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0
Justification: A literature review is missing. While references are provided for some standard results, they lack the required level of precision, e.g. citing a book without precise page locations. Multiple established results are reproved without proper attribution, e.g. in Section 8.4.

2 Submission with inline comments

3 The problem

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, be the closed Euclidean unit ball. For $f \in C^\alpha(D)$ with Hölder–Zygmund smoothness $\alpha > 1$ (and f bounded away from 0), let $u = u_f$ be the unique solution of Darcy's problem

$$\operatorname{div}(f\nabla u) = 1 \ \text{ on } D, \qquad u|_{\partial D} = 0.$$

We observe

$$Y_n = u_f + \tfrac{1}{\sqrt{n}}\,\mathbb{W},$$

where $\mathbb{W}$ is a Gaussian white noise on $L^2(D)$. For $r \in (0,1)$ let

$$\mathcal{F}(\alpha, r) = \{h \in C^\alpha(D) : \ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r\},$$

where $\mathbf{1}$ is the constant function 1. For small but fixed $r \in (0,1)$, the question is: what is the minimax rate, as $n \to \infty$ and up to logarithmic factors, of

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\big[\|\hat{f}_n - f_0\|^2_{L^2(D)}\big] \ ?$$

4 Result

Theorem 4.1. *Fix $\alpha > 1$ and $d \geq 2$.*

(i) ***(Lower bound.)*** *For every $r \in (0,1)$ there are $c > 0$ and n_0 such that for $n \geq n_0$*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\hat{f} - f\|^2_{L^2(D)} \ \geq \ c\, n^{-2\alpha/(2\alpha+d+3)}.$$

(ii) ***(Upper bound.)*** *There is $r_0 = r_0(d,\alpha) > 0$ such that for every $r \in (0, r_0]$ an explicit estimator $\tilde{f}$ (Definition 8.2) satisfies, for $n \geq n_0$,*

$$\sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\tilde{f} - f\|^2_{L^2(D)} \ \leq \ C(\log n)^4\, n^{-2\alpha/(2\alpha+d+3)}.$$

If α is not an integer, the factor $(\log n)^4$ can be dropped.

The constants r_0, c, C, n_0 depend only on d, α, r and on fixed auxiliary profiles and kernels chosen below.

Corollary 4.2 (Answer)**.** *For every fixed $r \in (0, r_0]$, up to logarithmic factors,*

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_n - f_0\|^2_{L^2(D)} \ \asymp \ n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

Everything in Sections 7–8 is argued here; the only external inputs are the standard facts collected, with references, in Section 6.

> **Reviewer 2**
>
> Literature review is missing.

5 Overview of the proof

Where $d+3$ comes from. Near $f = \mathbf{1}$ the solution is close to $u_1 := (|x|^2-1)/(2d)$, so $\nabla u_f \approx x/d$, a radial field with a single source at the origin. The linearised forward map $h \mapsto -L_f^{-1}\operatorname{div}(h\nabla u_f)$, with $L_f = \operatorname{div}(f\nabla\cdot)$ (Dirichlet), therefore smooths by order 2 for perturbations oscillating across the rays (in the $d-1$ angular directions), but only by order 1 along the rays, because $\operatorname{div}(h\,x/d)$ is a first-order operator along the rays. Heuristically, the symbol of the linearisation at $f = \mathbf{1}$ is $\asymp (1+|\xi_{\mathrm{rad}}|)/|\xi|^2$, and the exponent $d+3$ lies between the isotropic values $d+4$ and $d+2$. This heuristic only motivates the construction; it is not used in the proofs.

Lower bound. We perturb $\mathbf{1}$ by $M \asymp 2^{j(d-1)}$ disjoint bumps that oscillate at frequency 2^j in the angular directions only. Each has the form $2^{-j(\alpha+2)}b(|x|)\Delta\Phi_k$ with Φ_k homogeneous of degree 0; because of this form, the radial transport term $\operatorname{div}(h\,x/d)$ is again a Laplacian of a function of size $2^{-j(\alpha+2)}$. A duality argument then shows that the forward map shrinks each bump by 2^{-2j} in L^2. Balancing the Kullback–Leibler divergence with Assouad's lemma gives $2^j \asymp n^{1/(2\alpha+d+3)}$ and risk $2^{-2j\alpha}$.

Upper bound. We smooth the data at resolution 2^{-J}, $2^J \asymp n^{1/(2\alpha+d+3)}$, and estimate ∇u by $\hat V = \nabla K_J Y$. For r small, $\hat V$ is C^1-close to x/d, and the transport equation $\operatorname{div}(g\hat V) = 1$ has an explicit solution $T_{\hat V}^{-1}\mathbf{1}$, obtained by integrating along the backward flow of $\hat V$, which runs into a single sink. We linearise around the deterministic centring function $\bar f = T_{V_J}^{-1}\mathbf{1}$, where $V_J = \nabla K_J u$. This gives an exact decomposition: a bias $\bar f - f = O(J^2 2^{-J\alpha})$, a linear Gaussian term Z and a quadratic remainder. The variance of Z is the key point. Z integrates $\Delta\xi$ along rays, where ξ is the smoothed noise. Now $\Delta\xi$ has pointwise variance $n^{-1}2^{J(d+4)}$ and correlation length 2^{-J}, and integrating it along a ray of length $O(1)$ gains one factor 2^{-J}. Hence $\mathbb{E}\|Z\|_2^2 \lesssim n^{-1}2^{J(d+3)}$: the noise is smoothed by one order along the flow, and this produces $d+3$. The quadratic remainder is negligible because $\alpha > 1$.

6 Conventions and standard facts

C^s denotes the Hölder–Zygmund space $B^s_{\infty\infty}$; on D it consists of restrictions, with the quotient norm. Constants c, C may change from line to line. They never depend on j, J, n or on $f \in \mathcal{F}(\alpha, r)$. D° denotes the open unit ball. We write $u_1 := (|x|^2-1)/(2d)$, the solution for $f = \mathbf{1}$, so that $\nabla u_1 = x/d$. For $g \geq 1-r$ write $L_g = \operatorname{div}(g\nabla\cdot)$ with Dirichlet boundary conditions. Expectation is $\mathbb{E}$, and P_f is the law of Y_n. In the white-noise model, by the Cameron–Martin theorem (see e.g. [6]),

$$\mathrm{KL}(P_f, P_g) = \tfrac{n}{2}\|u_f - u_g\|_{L^2}^2.$$

We use the following known results without proof.

(E1) *Global H^2 regularity* [5, Theorem 8.12], combined with the energy estimate. Let $g \in C^{0,1}(D)$ with $g \geq 1-r$. The solution $w \in H_0^1(D)$ of $\operatorname{div}(g\nabla w) = F \in L^2$ satisfies $\|w\|_{H^2} \leq C\|F\|_2$, where C depends only on d, $1-r$ and an upper bound for $\|g\|_{C^{0,1}}$.

> **Reviewer 2**
>
> Inconsistent notation for the L^2-norm. Such issues persists throughout the document.

(E2) *$W^{2,p}$ theory, $1 < p < \infty$* [5, Theorem 9.15, Lemma 9.17]. Let $g \in C^1(D)$ with $g \geq 1 - r$. The Dirichlet problem $g\Delta w + \nabla g \cdot \nabla w = F \in L^p$, $w \in W^{2,p} \cap W^{1,p}_0$, has a unique solution, and $\|w\|_{W^{2,p}} \leq C\|F\|_p$. The constant is uniform over $\mathcal{F}(\alpha, r)$, which is bounded in C^α and hence precompact in C^1.

(GN) *Gagliardo–Nirenberg inequality on the ball* [4, 8]. For $p > d$ and $t = (1 + d/p)/2$, $\|\nabla w\|_\infty \leq C\|w\|^t_{W^{2,p}}\|w\|^{1-t}_p + C\|w\|_p$.

(As) *Assouad's lemma* [1]; see also [12, Ch. 2]. Let $\{P_\theta : \theta \in \{0,1\}^M\}$ be probability measures with $\mathrm{TV}(P_\theta, P_{\theta'}) \leq \tau < 1$ whenever θ, θ' differ in exactly one coordinate. Then for every estimator $\hat\theta$ with values in $\{0,1\}^M$, $\max_\theta \mathbb{E}_\theta \sum_k \mathbf{1}\{\hat\theta_k \neq \theta_k\} \geq \frac{M}{2}(1-\tau)$. We also use Pinsker's inequality $\mathrm{TV} \leq \sqrt{\mathrm{KL}/2}$ [12, Ch. 2].

(C1) *Seeley-type extension* [9]; see also [10]. There is a linear operator E with $Eg = g$ on D that is bounded $C^s(D) \to C^s(\mathbb{R}^d)$ for $0 < s \leq S_0$, where we fix $S_0 > \alpha + 2$. It has the form $Eg(t) = \chi(|t|)\sum_i a_i\, g(R_i t)$ for $|t| > 1$, where $R_i t = (1 - \lambda_i(|t| - 1))\, t/|t|$, finitely many $\lambda_i > 0$, and χ is a smooth cutoff vanishing for $|t| \geq 1 + \eta$, with $\lambda_i \eta < 1$.

(C2) *Kernel approximation* (see e.g. [6]). Let $k \in C^\infty_c(\mathbb{R}^d)$ with $\int k = 1$ and vanishing moments of orders $1, \ldots, N_0$, where we fix an integer $N_0 \geq 2\alpha + 2$. Put $k_J = 2^{Jd}k(2^J\cdot)$. Then $\|D^m(k_J * g - g)\|_\infty \leq C2^{-J(s-m)}\|g\|_{C^s}$ for integers $0 \leq m < s$ and $0 < s \leq N_0$.

(LP) *Littlewood–Paley theory* [10, 6]. For a fixed Littlewood–Paley decomposition $g = \sum_{j\geq 0}\Delta_j g$ we have $\|g\|_{C^s} \asymp \sup_j 2^{js}\|\Delta_j g\|_\infty$ for $s > 0$. Bernstein's inequality gives $\|D^m\Delta_j g\|_\infty \leq C2^{jm}\|\Delta_j g\|_\infty$. Moreover $C^s \subset C^{1,\beta_0}$ for $s > 1$, with $\beta_0 = \min(s - 1, 1/2)$.

(C3) *Taylor formula in Zygmund spaces; algebra property* (standard; see [10]). Let $\phi \in C^s$, $s > 0$, and $|h| \leq 1$. Then $\Delta_h\phi(x) := \phi(x + h) - \phi(x) = P_\phi(x, h) + r_\phi(x, h)$, where $P_\phi(x, h) = \sum_{1\leq|\iota|<s} \partial^\iota\phi(x)h^\iota/\iota!$ and $|r_\phi(x, h)| \leq C\|\phi\|_{C^s}|h|^s\Lambda(h)$ with $\Lambda(h) = 1+\log(1/|h|)$; the logarithm is needed only when s is an integer. Also $C^s \cdot C^s \subset C^s$, with $\|\phi\psi\|_{C^s} \leq C\|\phi\|_{C^s}\|\psi\|_{C^s}$.

(S) *Schauder estimate in Zygmund spaces* (standard). The Dirichlet Laplacian satisfies: $\Delta_D^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ is bounded for $\alpha - 1 > 0$. For non-integer α this is classical Schauder theory [5, Ch. 6].

(ODE) *C^k dependence of flows on initial data*, with the variational equations [7, Ch. V].

(G) *Gaussian facts* [6]. Let $y \mapsto \varphi_y \in L^2(D)$ be C^{m+1} on an open set, with locally Lipschitz derivatives. Then $\mathbb{W}(\varphi_y)$ has a C^m version with $\partial^\iota_y\mathbb{W}(\varphi_y) = \mathbb{W}(\partial^\iota_y\varphi_y)$ for $|\iota| \leq m$. Bochner integrals in L^2 commute with $\mathbb{W}$. We also use Dudley's entropy bound [3] and the Borell–Tsirelson–Ibragimov–Sudakov (Borell–TIS) inequality [2, 11].

7 The lower bound

Throughout this section $\alpha > 1$, $d \geq 2$ and $r \in (0, 1)$ are fixed, and $j \geq j_0$ is large.

7.1 The hypercube

Directions and bumps. Choose $\omega_1, \ldots, \omega_M \in S^{d-1}$ with pairwise geodesic distance $\geq 2 \cdot 2^{-j}$ and $M \geq c_d 2^{j(d-1)}$ (a maximal separated set). Fix $\varphi_0 \in C^\infty_c([0, 1/2))$ that is constant near 0 and not identically 0. Let $\Phi_k(\omega) = \varphi_0(2^j d_S(\omega, \omega_k))$, where d_S is the geodesic distance. The Φ_k are

smooth, are rotations of one another, and have pairwise disjoint supports, which are caps of radius $2^{-j}/2$.

Extension and scaling. Extend each Φ_k to $\mathbb{R}^d \setminus \{0\}$ as a function homogeneous of degree 0. Then

$$\Delta\Phi_k(x) = |x|^{-2}(\Delta_S\Phi_k)(x/|x|),$$

where Δ_S is the Laplace–Beltrami operator. Put $\chi_k := 2^{-2j}\Delta_S\Phi_k$. In rescaled normal coordinates this is the flat Laplacian of a fixed nonzero compactly supported bump, up to $O(2^{-j})$ metric corrections. Hence $\|\partial^m\chi_k\|_\infty \le C_m 2^{jm}$, and for $j \ge j_0$

$$c2^{-j(d-1)} \le \|\chi_k\|^2_{L^2(S^{d-1})} \le C2^{-j(d-1)}.$$

The lower bound holds because the Laplacian of a nonzero compactly supported function is nonzero, and the value does not depend on k by rotation invariance.

The perturbations. Fix $b \in C_c^\infty((1/4, 3/4))$, not identically 0. Define

$$h_k(x) := \kappa 2^{-j(\alpha+2)} b(|x|)\,\Delta\Phi_k(x) = \kappa 2^{-j\alpha} b(|x|)|x|^{-2}\chi_k(x/|x|), \qquad f_\theta := \mathbf{1} + \sum_{k=1}^{M} \theta_k h_k, \quad \theta \in \{0,1\}^M.$$

> **Reviewer 2**
>
> It would helpful to mention that f_θ are the alternatives used for the classical lower bound proof in the following.

Lemma 7.1. *There is $\kappa = \kappa(r) > 0$ such that $f_\theta \in \mathcal{F}(\alpha, r)$ for all θ and all j. The supports S_k of the h_k are pairwise disjoint, and*

$$\|h_k\|_2^2 \asymp 2^{-j(2\alpha+d-1)}, \qquad \|\Phi_k\|_{L^2(\{1/4<|x|<3/4\})} \asymp 2^{-j(d-1)/2}, \qquad \Big\|\sum_k \theta_k\Phi_k\Big\|_\infty \le \|\varphi_0\|_\infty.$$

Proof. Let $G_\theta = \sum_k \theta_k\chi_k$. Each point of the sphere has a neighbourhood meeting at most one support, so $\|D^m G_\theta\|_\infty \le C2^{jm}$ and $\|G_\theta\|_\infty \le C$. Take an integer $m > \alpha$. Then $|\Delta_h^m(2^{-j\alpha}G_\theta)| \le C2^{-j\alpha}\min((2^j|h|)^m, 1) \le C|h|^\alpha$. Hence $\|2^{-j\alpha}G_\theta\|_{C^\alpha(S^{d-1})} \le C$ uniformly in θ and j. Composing with $x \mapsto x/|x|$ and multiplying by the smooth function $b(|x|)|x|^{-2}$, supported in the annulus, keeps this bounded. Choosing κ small gives $\|f_\theta - \mathbf{1}\|_{C^\alpha(D)} \le r$. Disjointness of the supports is clear. In polar coordinates, $\|h_k\|_2^2 = \kappa^2 2^{-2j\alpha}\int b(\rho)^2\rho^{d-5}d\rho\,\|\chi_k\|^2_{L^2(S^{d-1})} \asymp 2^{-j(2\alpha+d-1)}$. The remaining two statements are immediate from the construction. □

By disjointness, $\|f_\theta - f_{\theta'}\|_2^2 = \sum_{k:\theta_k\neq\theta'_k}\|h_k\|_2^2$.

> **Reviewer 2**
>
> This is an important step in the proof of the lower bound, but is presented without context. This is a reoccurring issue.

7.2 The key estimate

> **Reviewer 2**
>
> An explanation of the goal of the subsections should be added (role within the lower bound proof). This is a reoccurring issue.

Proposition 7.2. *If $\theta, \theta' \in \{0,1\}^M$ differ only in coordinate k, then $\|u_{f_\theta} - u_{f_{\theta'}}\|_2 \le C2^{-2j}\|h_k\|_2$. Consequently $\|u_{f_\theta} - u_{f_{\theta'}}\|_2^2 \le C2^{-j(2\alpha+d+3)}$.*

Proof. Exact identity. Let $f = f_{\theta'}$, $g = f_\theta = f + h$ with $h = \pm h_k$, and $V = \nabla u_f$. Subtracting the two equations gives $\operatorname{div}(g\nabla(u_g - u_f)) = 1 - \operatorname{div}(g\nabla u_f) = -\operatorname{div}(hV)$. Hence

$$u_g - u_f = -L_g^{-1}\operatorname{div}(hV).$$

(i) Duality. L_g is self-adjoint on H_0^1, so $\|L_g^{-1}F\|_2 = \sup_{\|\phi\|_2\le 1}\langle F, \psi\rangle$ with $\psi := L_g^{-1}\phi$. Since $g \ge 1 - r$ and g is Lipschitz (as $\alpha > 1$), (E1) gives $\|\psi\|_{H^2} \le C$. It therefore suffices to show

$$|\langle \operatorname{div}(h_k V), \psi\rangle| \le C2^{-2j}\|h_k\|_2\|\psi\|_{H^2} \qquad \text{for } \psi \in H^2 \cap H_0^1. \tag{1}$$

Split $V = x/d + \nabla v$ with $v := u_f - u_1$.

(ii) Radial part. For $G = b(|x|)\Delta\Phi_k$ we claim

$$\operatorname{div}(G\,x/d) = b_1(|x|)\Delta\Phi_k, \qquad b_1(\rho) := \frac{\rho b'(\rho) + (d-2)b(\rho)}{d} \in C_c^\infty((1/4, 3/4)). \tag{2}$$

Indeed, $\Psi := \Delta\Phi_k$ is homogeneous of degree -2, so Euler's relation gives $x \cdot \nabla G = \rho b'\Psi - 2b\Psi$. Since $\operatorname{div}(G\,x) = x \cdot \nabla G + dG$, dividing by d gives (2). The function $b_1\psi$ has compact support in the annulus, where Φ_k is smooth. Integrating by parts twice and using Lemma 7.1,

$$|\langle \operatorname{div}(h_k\,x/d), \psi\rangle| = \kappa 2^{-j(\alpha+2)}\Big|\int \Phi_k\,\Delta(b_1\psi)\Big| \le C2^{-j(\alpha+2)}2^{-j(d-1)/2}\|\psi\|_{H^2} \le C2^{-2j}\|h_k\|_2\|\psi\|_{H^2}.$$

(iii) Drift part. Since h_k has compact support, $|\langle \operatorname{div}(h_k\nabla v), \psi\rangle| = |\int h_k\nabla v\cdot\nabla\psi| \le \|h_k\|_2\|\nabla v\|_\infty\|\psi\|_{H^1}$. It therefore suffices to show $\|\nabla v\|_\infty \le C2^{-j(\alpha+1-d/p)}$ for every $p > d$, and then to take $p > d$ with $d/p \le \alpha - 1$.

Equation for v. We have $\operatorname{div}(f\nabla v) = 1 - \operatorname{div}(f\,x/d) = -\operatorname{div}((f-1)x/d) =: -q$. By (2) applied termwise, $q = \kappa 2^{-j(\alpha+2)}b_1(|x|)\Delta\Phi_{\theta'}$ with $\Phi_{\theta'} = \sum_l \theta'_l\Phi_l$.

$W^{2,p}$ bound. Since $q = \kappa 2^{-j\alpha}b_1(|x|)|x|^{-2}\sum_l \theta'_l\chi_l(x/|x|)$ and the χ_l are bounded with disjoint supports, $\|q\|_p \le C2^{-j\alpha}$. Hence $\|v\|_{W^{2,p}} \le C2^{-j\alpha}$ by (E2).

L^p bound. Take ϕ with $\|\phi\|_{p'} \le 1$ and let $\psi = L_f^{-1}\phi \in W^{2,p'} \cap W_0^{1,p'}$, so that $\|\psi\|_{W^{2,p'}} \le C$ by (E2). By Green's identity, and integrating by parts twice as in (ii),

$$\Big|\int v\phi\Big| = \Big|\int q\psi\Big| = \kappa 2^{-j(\alpha+2)}\Big|\int \Phi_{\theta'}\,\Delta(b_1\psi)\Big| \le C2^{-j(\alpha+2)}\|\varphi_0\|_\infty\|\psi\|_{W^{2,p'}} \le C2^{-j(\alpha+2)}.$$

So $\|v\|_p \le C2^{-j(\alpha+2)}$.

Interpolation. By (GN), $\|\nabla v\|_\infty \le C2^{-j\alpha t}2^{-j(\alpha+2)(1-t)} + C2^{-j(\alpha+2)} \le C2^{-j(\alpha+1-d/p)}$, since $2(1-t) = 1 - d/p$.

Combining (ii) and (iii) proves (1), hence the first claim. The second follows from $\|h_k\|_2^2 \le C2^{-j(2\alpha+d-1)}$. □

7.3 Conclusion via Assouad's lemma

Proof of Theorem 4.1(i). By Proposition 7.2, $\mathrm{KL}(P_{f_\theta}, P_{f_{\theta'}}) \le Cn2^{-j(2\alpha+d+3)}$ for neighbours θ, θ'. Choose $2^j \asymp n^{1/(2\alpha+d+3)}$, with a constant large enough that KL $\le 1/8$ for all neighbours. Pinsker's inequality then gives TV $\le 1/4$.

Given any estimator $\hat f$, let $\hat\theta_k$ minimise $\|(\hat f - 1 - th_k)\mathbf{1}_{S_k}\|_2$ over $t \in \{0,1\}$. On S_k we have $f_\theta = 1 + \theta_k h_k$. If $\hat\theta_k \ne \theta_k$, the triangle inequality gives $\|h_k\|_2 \le \|(\hat f - 1 - \hat\theta_k h_k)\mathbf{1}_{S_k}\|_2 + \|(\hat f - 1 - \theta_k h_k)\mathbf{1}_{S_k}\|_2 \le 2\|(\hat f - f_\theta)\mathbf{1}_{S_k}\|_2$. Since the S_k are disjoint,

$$\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \sum_k \|h_k\|_2^2 \mathbf{1}\{\hat\theta_k \ne \theta_k\}.$$

By (As) with $\tau = 1/4$ and Lemma 7.1,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\hat f - f\|_2^2 \ge \max_\theta \mathbb{E}_\theta\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \min_k \|h_k\|_2^2 \cdot \tfrac{M}{2}\cdot\tfrac34 \ge c2^{j(d-1)}2^{-j(2\alpha+d-1)} = c2^{-2j\alpha} \asymp n^{-2\alpha/(2\alpha+d+3)}.$$

The condition $\alpha > 1$ was used only in (iii), and to make $f \in C^1$ for (E1) and (E2). □

8 The upper bound

Throughout, $\alpha > 1$ and $d \ge 2$ are fixed, $f \in \mathcal{F}(\alpha, r)$, and $r \le r_0$, where r_0 is reduced finitely many times below.

> **Reviewer 2**
>
> For improved exposition, the following technical subsections should be moved after the definition and analysis of the estimator.

8.1 A Schauder bound

Lemma 8.1. *There is $r_0 = r_0(d,\alpha)$ such that for $r \le r_0$ and all $f \in \mathcal{F}(\alpha, r)$,*

$$\|u_f - u_1\|_{C^{\alpha+1}(D)} \le Cr, \qquad \|u_f\|_{C^{\alpha+1}(D)} \le C. \tag{3}$$

Proof. Write $L_f = \Delta + \operatorname{div}((f-1)\nabla\cdot)$. By the algebra property (C3), $\|\operatorname{div}((f-1)\nabla v)\|_{C^{\alpha-1}} \le C_1\|f-1\|_{C^\alpha}\|v\|_{C^{\alpha+1}} \le C_1 r\|v\|_{C^{\alpha+1}}$. Take r_0 with $C_1 r_0\|\Delta_D^{-1}\| \le 1/2$. Writing $L_f v = F$ as $(I + \Delta_D^{-1}\operatorname{div}((f-1)\nabla\cdot))v = \Delta_D^{-1}F$ on the closed subspace of $C^{\alpha+1}(D)$ with zero trace, a Neumann series using (S) shows that $L_f^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ (Dirichlet data) has norm $\le 2\|\Delta_D^{-1}\|$. The solution so obtained is classical, hence equal to the weak solution. As in Proposition 7.2, $u_f - u_1 = -L_f^{-1}\operatorname{div}((f-1)x/d)$, and the right-hand side has $C^{\alpha-1}$ norm $\le Cr$. □

This is the first place where smallness of r enters.

8.2 Smoothing, noise and the estimator

Resolution and smoother. Let $J := \lfloor \log_2 n/(2\alpha+d+3)\rfloor$. Then

$$n^{-1/2} \le 2^{-J(\alpha+(d+3)/2)}, \qquad n^{-1}2^{J(d+3)} \le 2^{-2J\alpha}, \qquad 2^{-2J\alpha} \le 2^{2\alpha}n^{-2\alpha/(2\alpha+d+3)}, \qquad J \le \log_2 n. \tag{4}$$

Put $K_J g := (k_J * Eg)|_D$, with k_J from (C2) and E from (C1). On data, $K_J Y(x) = \langle Y, k_J(x,\cdot)\rangle$, where $k_J(x,\cdot) := E^*[k_J(x-\cdot)] \in L^2(D)$; this is defined for all $x \in \mathbb{R}^d$. By the change of variables $z = R_i t$, E^* is the identity on D plus finitely many terms of the form $a_i(\chi\varphi)(R_i^{-1}z)|\det DR_i^{-1}(z)|$. These are supported on shells $\{1-\lambda_i\eta < |z| < 1\}$ and have bounded Jacobians. This gives the following kernel bounds.

(K1) $|\partial_x^\iota k_J(x,z)| \le C2^{J(d+|\iota|)}$ for $|\iota| \le 4$.

(K2) $\partial_x^\iota k_J(x,z) = 0$ unless $|x-z| \le C_0 2^{-J}$. Indeed, for the reflected terms let $t = R_i^{-1}z$, so $|t| > 1$. If $x \in D$ and $|x-t| \le C2^{-J}$, then $|t|-1 \le C2^{-J}$ and $|z-t| = (1+\lambda_i)(|t|-1) \le C2^{-J}$.

(K3) $\|\partial_x^\iota k_J(x,\cdot)\|_2^2 \le C2^{J(d+2|\iota|)}$, and $x \mapsto \partial_x^\iota k_J(x,\cdot)$ is Lipschitz into L^2 with constant $C2^{J(d/2+|\iota|+1)}$, for $|\iota| \le 3$. This follows from (K1), (K2) and the fact that the support in z has volume $\le C2^{-Jd}$.

Notation. Let $U = Eu$, $F = Ef$, $V = \nabla u$ and $V^E = \nabla U$. By (C1) and (3), $\|F\|_{C^\alpha} + \|V^E\|_{C^\alpha} \le C$. Let

$$V_J := \nabla K_J u = (k_J * V^E)|_D, \qquad e := K_J u - u, \qquad \xi := n^{-1/2} K_J \mathbb{W}, \qquad \hat{V} := \nabla K_J Y = V_J + \nabla\xi.$$

V_J is the restriction of a C^∞ field on $\mathbb{R}^d$. By (G) and (K1)–(K3), ξ is a centred Gaussian field with C^3 sample paths on a neighbourhood of D (it is in fact smooth), and ξ does not depend on f.

Good event. Let

$$\mathcal{A} := \Big\{ \max_{|\iota|=m} \sup_D |\partial^\iota \xi| \le Jn^{-1/2}2^{J(d/2+m)},\ m = 1,2,3 \Big\}.$$

On $\mathcal{A}$, by (4) and after adjusting constants for operator norms,

$$\|\nabla\xi\|_\infty \le CJ2^{-J(\alpha+1/2)}, \qquad \|D^2\xi\|_\infty \le CJ2^{-J(\alpha-1/2)}, \qquad \|D^3\xi\|_\infty \le CJ2^{-J(\alpha-3/2)}. \tag{5}$$

Definition 8.2 (Estimator)**.** Let $\varepsilon_0 := 1/(8d)$, and define the observable event $\mathcal{B} := \{\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon_0\}$. On $\mathcal{B}$ set $\hat{f} := T_{\hat{V}}^{-1}\mathbf{1}$ (Definition 8.6); off $\mathcal{B}$ set $\hat{f} := \mathbf{1}$. Let $\tilde{f} := \min(\max(\hat{f}, 1-r), 1+r)$.

> **Reviewer 2**
>
> This is a forward reference to Definition 8.6. The definition of the estimator should be briefly motivated by the inversion formula of the Darcy flow.

Since $\|f - 1\|_\infty \le r$, clipping can only decrease $|\tilde{f} - f|$ pointwise. The estimator uses only Y, n, α, r and the fixed kernels. The event $\mathcal{A}$ itself is not observable: ξ does not depend on f, but $\xi = K_J Y - K_J u$ cannot be computed from the data because u is unknown. This is why the estimator is defined on $\mathcal{B}$; Lemma 8.4 shows $\mathcal{A} \subset \mathcal{B}$.

8.3 The smoothed fields

Lemma 8.3. *Let $g \in C^s(\mathbb{R}^d)$, $s > 0$, and let $m \ge 1$ be an integer. Then $\|D^m(k_J * g)\|_\infty$ is at most $C2^{J(m-s)}\|g\|_{C^s}$ if $m > s$, at most $C(J+1)\|g\|_{C^s}$ if $m = s$, and at most $C\|g\|_{C^s}$ if $m < s$.*

Proof. By (LP), $D^m(k_J * g) = \sum_j (D^m k_J) * \Delta_j g$, and the series converges uniformly because each term is $\le \|D^m k_J\|_1 \|\Delta_j g\|_\infty \le C2^{Jm}2^{-js}\|g\|_{C^s}$. For $j \le J$, write the term as $k_J * D^m\Delta_j g$; by Bernstein it is $\le C2^{j(m-s)}\|g\|_{C^s}$. For $j > J$, the first bound gives $C2^{Jm-js}\|g\|_{C^s}$, and summing over $j > J$ gives $C2^{J(m-s)}\|g\|_{C^s}$. Summing the bound $C2^{j(m-s)}$ over $j \le J$ gives the three cases. □

Lemma 8.4. *Let $\varepsilon \le \varepsilon_0$. There are r_0 and J_0 such that for $r \le r_0$ and $J \ge J_0$:*

(1) $\|V_J - x/d\|_{C^1(D)} \le \varepsilon/2$;

(2) on $\mathcal{A}$, $\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon$; in particular (with $\varepsilon = \varepsilon_0$) $\mathcal{A} \subset \mathcal{B}$;

(3) $\|D^m V_J\|_\infty \le C J 2^{J(m-\alpha)_+}$ for $m = 2, 3$, with no factor J unless $\alpha = m$.

Proof. Item (3) is Lemma 8.3 with $g = V^E$ and $s = \alpha$. For (1), on D we have $V_J - x/d = (k_J * V^E - V^E) + \nabla(u - u_1)$. By (LP), $D^i V^E$ is β_0-Hölder for $i = 0, 1$, with $\beta_0 = \min(\alpha - 1, 1/2)$. Hence $|D^i(k_J * V^E - V^E)| \le C \int |k_J(y)||y|^{\beta_0} dy \le C 2^{-J\beta_0}$. By (3), $\|\nabla(u - u_1)\|_{C^1} \le Cr$. Take r_0 and J_0 such that $C(2^{-J_0\beta_0} + r_0) \le \varepsilon/2$. For (2), on $\mathcal{A}$, (5) gives $\|\nabla\xi\|_{C^1} \le C J 2^{-J(\alpha-1/2)} \le \varepsilon/2$ for $J \ge J_0$. □

8.4 Geometry of the flow and the transport inverse

> **Reviewer 2**
>
> Established theory with very brief proofs without attribution to previous work.

Setting. Let X be the restriction to D of a C^k field ($k \ge 1$) on a neighbourhood of D, with

$$\|X - x/d\|_{C^1(D)} := \max\big(\sup_D |X - x/d|,\ \sup_D \|DX - I/d\|\big) \le \varepsilon \le 1/(8d).$$

Put $\nu := 1/d - \varepsilon \ge 7/(8d)$. Let Φ_t be the flow of X, let $\gamma(s) = \gamma_x(s) = \Phi_{-s}x$ be the backward ray from x, and let

$$w(s, x) := \exp\Big(-\int_0^s \operatorname{div} X(\Phi_{-\sigma}x)\, d\sigma\Big).$$

Lemma 8.5. *In this setting the following hold.*

(1) X has exactly one zero $x_X \in D$, and $|x_X| \le d\varepsilon$.

(2) With $\rho(y) = |y - x_X|$: $(y - x_X) \cdot X(y) \ge (1/d - \varepsilon)\rho^2$ and $|X(y)| \le (1/d + \varepsilon)\rho$ on D.

(3) For $x \in D$, $\Phi_{-s}x$ exists for all $s \ge 0$ and stays in D. It lies in D° for $s > 0$, and D° is invariant.

(4) $\varrho(s) := \rho(\gamma(s))$ satisfies $\varrho(0)e^{-(1/d+\varepsilon)s} \le \varrho(s) \le \varrho(0)e^{-(1/d-\varepsilon)s}$. If $\varrho(0) > 0$, then $-(1/d + \varepsilon)\varrho \le \varrho' \le -\nu\varrho$.

(5) The propagator $P(s, \sigma)$ of $y' = -DX(\gamma)y$ satisfies $\|P(s, \sigma)\| \le e^{-\nu(s-\sigma)}$. In particular $\|D\Phi_{-s}\| \le e^{-\nu s}$.

(6) $1 - d\varepsilon \le \operatorname{div} X \le 1 + d\varepsilon$, so $e^{-(1+d\varepsilon)s} \le w \le e^{-(1-d\varepsilon)s} \le e^{-7s/8}$.

Proof. (1) The map $x \mapsto x - dX(x) = -d(X - x/d)$ sends D into $B(0, d\varepsilon) \subset D$ and is a $d\varepsilon$-contraction on the convex set D. Its unique fixed point is the unique zero of X. (2) $X(y) = Q(y)(y - x_X)$ with $Q = \int_0^1 DX(x_X + t(y - x_X))\, dt$ and $\|Q - I/d\| \le \varepsilon$; the segment lies in D by convexity. (3) For $|z| = 1$, $z \cdot X(z) \ge 1/d - \varepsilon > 0$, so $\frac{d}{ds}|\gamma|^2 = -2\gamma \cdot X(\gamma) < 0$ whenever $|\gamma| = 1$. A first-exit-time argument gives invariance of D and of D°, and shows that boundary points enter D° immediately. Global existence holds because D is compact. (4) $(\varrho^2)' = -2(\gamma - x_X) \cdot X(\gamma)$; apply (2) and Gronwall's inequality. (5) $v \cdot DX\, v \ge \nu|v|^2$ on D, so $(|y|^2)' \le -2\nu|y|^2$. Also $D\Phi_{-s} = P(s, 0)$. (6) $\operatorname{div} X = 1 + \operatorname{tr}(DX - I/d)$ and $|\operatorname{tr} A| \le d\|A\|$. □

Definition 8.6. For $g \in C^1(D)$ put $T_X g := \operatorname{div}(gX)$. For $H \in C(D)$ put

$$T_X^{-1} H(x) := \int_0^\infty H(\Phi_{-s}x)\, w(s, x)\, ds.$$

> **Reviewer 2**
>
> Overly complicated notation.

By Lemma 8.5(6) the integral converges absolutely and uniformly, and $T_X^{-1}H \in C(D)$.

Since $w(s, \Phi_{-t}x) = w(s+t, x)/w(t, x)$, we have the semigroup identity

$$w(t,x)\, T_X^{-1}H(\Phi_{-t}x) = \int_t^\infty H(\Phi_{-s}x)\, w(s,x)\, ds. \tag{6}$$

Lemma 8.7. *In the setting of Lemma 8.5:*

(a) (Uniqueness.) *If $g \in C^1(D)$ and $\operatorname{div}(gX) = H$ on D° with $H \in C(D)$, then $g = T_X^{-1}H$.*

(a′) (Right inverse.) *If $H \in C(D)$ and $g := T_X^{-1}H \in C^1(D)$, then $\operatorname{div}(gX) = H$ on D.*

(b) $\|T_X^{-1}H\|_\infty \le \frac{8}{7}\|H\|_\infty \le 2\|H\|_\infty$.

(c) (Boundary layer.) *If $H = 0$ on $\{|x| \le 1-\delta\}$, then $\|T_X^{-1}H\|_\infty \le 6d\delta\|H\|_\infty$.*

> **Reviewer 2**
>
> Bound-properties of T_X^{-1} are split across Lemmas 8.7 and 8.9 and should be presented together.

Proof. (a) Fix $x \in D$. For $s > 0$, $\gamma(s) \in D^\circ$, and $\psi(s) = g(\gamma(s))w(s,x)$ satisfies $\psi' = -(X\cdot\nabla g + g\operatorname{div} X)(\gamma)\, w = -H(\gamma)\, w$. Integrate over $[\delta, S]$, then let $\delta \to 0$ (by continuity) and $S \to \infty$ (using $|\psi(S)| \le \|g\|_\infty e^{-7S/8}$). This gives $g(x) = \int_0^\infty H(\gamma(s))w(s,x)\, ds$.

(a′) Fix $x \in D^\circ$ and differentiate (6) at $t = 0^+$. The right side gives $-H(x)$. Since $w(0,x) = 1$ and $\partial_t w(0,x) = -\operatorname{div} X(x)$, the left side gives $-\nabla g(x)\cdot X(x) - g(x)\operatorname{div} X(x)$. Extend to D by continuity.

(b) Use $w \le e^{-7s/8}$.

(c) The case $\delta > 1/2$ follows from (b), since then $6d\delta > 2$. Let $\delta \le 1/2$ and $\phi = |\gamma|^2$. Where $|\gamma| \ge 1/2$, $\gamma\cdot X(\gamma) \ge |\gamma|(|\gamma|/d - \varepsilon) \ge 3/(16d)$, so $\phi' \le -3/(8d)$. Hence ϕ drops below $(1-\delta)^2$ within time $\le (1-(1-\delta)^2)\cdot 8d/3 \le 6d\delta$, and by the same sign argument it never returns. Since $H(\gamma(s)) = 0$ afterwards and $w \le 1$, the claim follows. □

Lemma 8.8 (Flow derivatives, uniform in s)**.** *Let $M = D\Phi_{-s}$, $N = D^2\Phi_{-s}$ and $\ell = \log w$, all differentiated in x.*

(1) If $X \in C^2$: $\|N(s)\| \le \|D^2X\|_\infty e^{-\nu s}/\nu$ and $|D_x\ell| \le \|D\operatorname{div} X\|_\infty/\nu$.

(2) If $X \in C^3$: $\|D_x^2\ell\| \le \|D^2\operatorname{div} X\|_\infty/(2\nu) + \|D\operatorname{div} X\|_\infty\|D^2X\|_\infty/\nu^2$.

Proof. N solves $N' = -DX(\gamma)N - D^2X(\gamma)[M\cdot, M\cdot]$ with $N(0) = 0$. By Duhamel's formula and Lemma 8.5(5), $\|N(s)\| \le \int_0^s e^{-\nu(s-\sigma)}\|D^2X\|_\infty e^{-2\nu\sigma}d\sigma \le \|D^2X\|_\infty e^{-\nu s}/\nu$. Next, $D_x\ell[v] = -\int_0^s D\operatorname{div} X(\gamma)[Mv]\, d\sigma$, and $\int_0^\infty e^{-\nu\sigma}d\sigma = 1/\nu$. Finally, $D_x^2\ell[v,v'] = -\int_0^s \big(D^2\operatorname{div} X(\gamma)[Mv, Mv'] + D\operatorname{div} X(\gamma)[N[v,v']]\big)d\sigma$; bound this with the two previous estimates. □

Differentiation under the integral. Let the integrand be $H(\Phi_{-s}x)w(s,x)$ or $w(s,x)$. By (ODE), and since the flow is defined on a neighbourhood of D, its x-derivatives up to the relevant order are jointly continuous on $[0,\infty)\times D$. By Lemma 8.8 and the identities $D_x w = w\, D_x\ell$ and $D_x^2 w = w(D_x^2\ell + D_x\ell \otimes D_x\ell)$, they are dominated by $Ke^{-7s/8}$. Truncating the integral at S, differentiating, and passing to the uniform limit gives the following.

Lemma 8.9. *All constants below depend only on d, uniformly over $\|X - x/d\|_{C^1} \le \varepsilon \le 1/(8d)$.*

(d) *If $X \in C^2$ and $H \in C^1(D)$, then $T_X^{-1}H \in C^1(D)$ and $\|\nabla T_X^{-1}H\|_\infty \le C_d(\|\nabla H\|_\infty + \|H\|_\infty \|D\operatorname{div} X\|_\infty)$.*

(e) *If $X \in C^3$, then $T_X^{-1}\mathbf{1} \in C^2(D)$, $\|\nabla T_X^{-1}\mathbf{1}\|_\infty \le C_d\|D\operatorname{div} X\|_\infty$, and $\|D^2 T_X^{-1}\mathbf{1}\|_\infty \le C_d\big(\|D^2\operatorname{div} X\|_\infty + \|D\operatorname{div} X\|_\infty^2 + \|D\operatorname{div} X\|_\infty\|D^2 X\|_\infty\big)$.*

8.5 The centring function and an exact decomposition

Fix $\varepsilon = \varepsilon_0$ in Lemma 8.4, so that Lemmas 8.5–8.9 apply to $X = V_J$ (with $\varepsilon_0/2$), and on $\mathcal{A}$ also to $X = \hat V$. Let

$$\bar f := T_{V_J}^{-1}\mathbf{1}.$$

Using $\|D^m \operatorname{div} V_J\|_\infty \le d\|D^{m+1}V_J\|_\infty$, Lemma 8.4(3) and Lemma 8.9(e),

$$\|\bar f\|_\infty \le 2, \qquad \|\nabla \bar f\|_\infty \le CJ2^{J(2-\alpha)_+}, \qquad \|D^2\bar f\|_\infty \le CJ^2 2^{J(3-\alpha)_+}. \tag{7}$$

The last bound uses $2(2-\alpha)_+ \le (3-\alpha)_+$ for $\alpha \ge 1$. Moreover $\bar f \in C^2(D)$, and $\operatorname{div}(\bar f V_J) = 1$ by Lemma 8.7(a′).

Noise term. Let $G_\xi := \operatorname{div}(\bar f\nabla\xi) = \bar f\Delta\xi + \nabla\bar f\cdot\nabla\xi \in C^1(D)$ and $Z := T_{V_J}^{-1}G_\xi$. Since V_J is deterministic, Z is defined for every realisation. By Lemma 8.9(d), $Z \in C^1(D)$, and $T_{V_J}Z = G_\xi$ by Lemma 8.7(a′).

Proposition 8.10. *On $\mathcal{A}$ (so that $\hat V$ satisfies the hypotheses of Lemma 8.5 and $\hat f = T_{\hat V}^{-1}\mathbf{1}$),*

$$\hat f - f = (\bar f - f) - Z + T_{\hat V}^{-1}\operatorname{div}(Z\nabla\xi). \tag{8}$$

> **Reviewer 2**
> Since the error decomposition is a key ingredient for establishing the upper bound, it should be explained in more detail. The dependence of $\hat f$ and $\bar f$ on J should be reflected in the notation.

Proof. $\hat V$ is C^3 on a neighbourhood of D, so $\hat f \in C^2$ by Lemma 8.9(e), and $T_{\hat V}\hat f = 1$

> **Reviewer 2**
> 1 should be **1**.

by Lemma 8.7(a′). Hence $T_{\hat V}(\hat f - \bar f) = 1 - \operatorname{div}(\bar f V_J) - \operatorname{div}(\bar f\nabla\xi) = -G_\xi$. Also $T_{\hat V}Z = T_{V_J}Z + \operatorname{div}(Z\nabla\xi) = G_\xi + \operatorname{div}(Z\nabla\xi)$. Therefore $T_{\hat V}(\hat f - \bar f + Z) = \operatorname{div}(Z\nabla\xi)$, which is continuous. Since $\hat f - \bar f + Z \in C^1$, Lemma 8.7(a) with $X = \hat V$ gives $\hat f - \bar f + Z = T_{\hat V}^{-1}\operatorname{div}(Z\nabla\xi)$, which is (8). □

8.6 The bias: $\|\bar f - f\|_\infty \le CJ^2 2^{-J\alpha}$

Proposition 8.11. *Uniformly over $f \in \mathcal{F}(\alpha, r)$, $\|\bar f - f\|_\infty \le CJ^2 2^{-J\alpha}$, with no factor J^2 when $\alpha \notin \mathbb{N}$.*

Proof. Reduction. $f \in C^1$ (as $\alpha > 1$), $u \in C^{\alpha+1} \subset C^2$ by (3), $\operatorname{div}(fV) = 1$, and $V_J - V = \nabla e$ on D. So $T_{V_J}(\bar f - f) = 1 - \operatorname{div}(fV_J) = -\operatorname{div}(f\nabla e)$, and by Lemma 8.7(a), $\bar f - f = -T_{V_J}^{-1}\operatorname{div}(f\nabla e)$.

Commutator. Let $g := F - k_J * F$ and $R(x) := -\int k_J(-h)\Delta_h F(x)\Delta_h V^E(x)\,dh$. Then

$$F\,(k_J * V^E) = k_J * (FV^E) + gV^E + R.$$

Indeed, $F(x)(k_J * V^E)(x) - k_J * (FV^E)(x) = \int k_J(y)[F(x) - F(x-y)]V^E(x-y)\,dy$. Insert $F(x) - F(x-y) = -\Delta_{-y}F(x)$ and $V^E(x-y) = V^E(x) + \Delta_{-y}V^E(x)$, and use $\int k_J = 1$; the first part gives gV^E and the second, after $h = -y$, gives R. Let $\Gamma := \operatorname{div}(FV^E) - 1$, which lies in $C^{\alpha-1}(\mathbb{R}^d)$ by the algebra property and vanishes on D. Since $\operatorname{div}(k_J * (FV^E)) = k_J * \Gamma + 1$, on D

$$\operatorname{div}(f\nabla e) = \operatorname{div}(fV_J) - 1 = k_J * \Gamma + \operatorname{div}(gV) + \operatorname{div} R.$$

We treat the three terms in turn, using linearity of $T_{V_J}^{-1}$.

(i) The term $\operatorname{div}(gV)$. Since $V = V_J - \nabla e$, we have $\operatorname{div}(gV) = T_{V_J} g - \operatorname{div}(g\nabla e)$. As $g \in C^1$, Lemma 8.7(a) gives $T_{V_J}^{-1}\operatorname{div}(gV) = g - T_{V_J}^{-1}\operatorname{div}(g\nabla e)$. By (C2) (applied to $F \in C^\alpha$ and $U \in C^{\alpha+1}$): $\|g\|_\infty \le C2^{-J\alpha}$, $\|\nabla g\|_\infty \le C2^{-J(\alpha-1)}$, $\|\nabla e\|_\infty \le C2^{-J\alpha}$ and $\|D^2 e\|_\infty \le C2^{-J(\alpha-1)}$. By Lemma 8.7(b), the contribution is $\le C2^{-J\alpha} + C2^{-J(2\alpha-1)}$.

(ii) The term $\operatorname{div} R$. We have $\partial R = -\int k_J(-h)\big[\Delta_h \partial F\,\Delta_h V^E + \Delta_h F\,\Delta_h \partial V^E\big]dh$. Each product has the form $\Delta_h\phi_1\,\Delta_h\phi_2$ with $\phi_i \in C^{s_i}$ and $(s_1, s_2) = (\alpha - 1, \alpha)$ or $(\alpha, \alpha - 1)$. By (C3), and using $|P_{\phi_i}| \le C|h|$ for $|h| \le 1$, the product equals a polynomial in h (with coefficients depending on x) whose monomials have degrees in $[2, s_1 + s_2) = [2, 2\alpha - 1)$. The remainder is bounded by $C|h|^{\min(1+s_1,\,1+s_2,\,s_1+s_2)}\Lambda(h)^2 = C|h|^\alpha\Lambda(h)^2$. When $s_i < 1$, P_{ϕ_i} is empty. The polynomial terms integrate to 0 against $k_J(-h)$, because $N_0 \ge 2\alpha + 2$. Since k_J is supported in $|h| \le C2^{-J}$, the substitution $h = 2^{-J}y$ gives $\int |k_J(h)||h|^\alpha\Lambda(h)^2 dh \le CJ^2 2^{-J\alpha}$. So $\|\operatorname{div} R\|_\infty \le CJ^2 2^{-J\alpha}$, with no J^2 for non-integer α (then no logarithm appears in (C3)). By Lemma 8.7(b), the contribution is $\le CJ^2 2^{-J\alpha}$.

(iii) The term $k_J * \Gamma$. On D, $k_J * \Gamma = k_J * \Gamma - \Gamma$, so by (C2) with $s = \alpha - 1$ and $m = 0$ it is $O(2^{-J(\alpha-1)})$. By the support of k_J, it vanishes at points of D at distance $> C2^{-J}$ from ∂D. By Lemma 8.7(c) with $\delta = C2^{-J}$, the contribution is $\le C2^{-J}2^{-J(\alpha-1)} = C2^{-J\alpha}$.

Adding (i)–(iii) gives the claim. □

8.7 The variance: $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$

Proposition 8.12. $\mathbb{E}\|Z\|_{L^2(D)}^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$.

Proof. Representation. Let $\mathcal{D}_y := \bar f(y)\Delta_y + \nabla\bar f(y)\cdot\nabla_y$. By (G), $G_\xi(y) = n^{-1/2}\mathbb{W}(\mathcal{D}_y k_J(y, \cdot))$, and since Bochner integrals commute with $\mathbb{W}$,

$$Z(x) = n^{-1/2}\mathbb{W}(h_x), \qquad h_x = \int_0^\infty w(s,x)\,\mathcal{D}_y k_J(y,\cdot)\big|_{y=\gamma_x(s)}\,ds,$$

where γ_x is the backward ray of V_J. Hence $\mathbb{E}Z(x)^2 = n^{-1}\|h_x\|_2^2$.

Kernel bound. Since $(2-\alpha)_+ < 1$, (7) gives $\|\nabla \bar f\|_\infty \le CJ2^{J(2-\alpha)_+} \le C2^J$. With (K1) and (K2), $|\mathcal{D}_y k_J(y,z)| \le C2^{J(d+2)}\mathbf{1}\{|y-z| \le \delta_J\}$, where $\delta_J := C_0 2^{-J}$. Let $\mu_x := \int_0^\infty w(s,x)\,\delta_{\gamma_x(s)}\,ds$, a measure on D of mass $\le 8/7$. Then

$$|h_x(z)| \le C2^{J(d+2)}\mu_x(B(z,\delta_J)), \qquad \|h_x\|_2^2 \le C2^{2J(d+2)}\int \mu_x(B(z,\delta_J))^2 dz \le C2^{2J(d+2)}\delta_J^d \int \mu_x(B(y,2\delta_J))\,\mu_x(dy).$$

The last step is Fubini: $\int \mu(B(z,\delta))^2 dz = \iint |B(y,\delta)\cap B(y',\delta)|\,\mu(dy)\mu(dy')$, and the intersection is empty unless $|y-y'| \le 2\delta$.

Ray measure. Let x_J be the zero of V_J, $\rho = |\cdot - x_J|$, and $\rho_x = \rho(x) > 0$. Let $\varrho(s) = \rho(\gamma_x(s))$. By Lemma 8.5(4),(6), $w(s,x) \le (\varrho(s)/\rho_x)^\beta$ with $\beta = d(1-d\varepsilon)/(1+d\varepsilon) \ge 7d/9 > 1$. Moreover ϱ decreases strictly from ρ_x to 0, with $|\varrho'| \ge \nu\varrho$. Changing variables $t = \varrho(s)$, for every $\varphi \ge 0$,

$$\int \varphi\, d\mu_x \le \frac{1}{\nu}\int_0^{\rho_x} \varphi(\gamma_x \text{ at radius } t)\,(t/\rho_x)^\beta\,\frac{dt}{t}. \tag{9}$$

Small-ball mass. If $|\gamma_x(s)-y| \le 2\delta_J$, then $|\varrho(s)-\rho(y)| \le 2\delta_J$. Hence, by (9), $m(y) := \mu_x(B(y,2\delta_J))$ satisfies

$$m(y) \le C\delta_J\rho(y)^{\beta-1}\rho_x^{-\beta} \text{ if } \rho(y) \ge 4\delta_J, \qquad m(y) \le C\min(1,\delta_J/\rho_x)^\beta \text{ if } \rho(y) < 4\delta_J.$$

(In the first case $t \in [\rho(y)/2, 3\rho(y)/2]$ ranges over an interval of length $4\delta_J$; in the second $t \le \min(6\delta_J, \rho_x)$.)

Integration. Apply (9) once more, with $\varphi = m$. The part with $t \ge 4\delta_J$ is $\le C\delta_J\rho_x^{-2\beta}\int_0^{\rho_x} t^{2\beta-2}dt \le C\delta_J/\rho_x$, since $2\beta - 1 > 0$. The part with $t < 4\delta_J$ is $\le C\min(1,\delta_J/\rho_x)^{2\beta} \le C\min(1,\delta_J/\rho_x)$. Trivially, $\int m\,d\mu_x \le \mu_x(D)^2 \le 4$. Hence $\int m\,d\mu_x \le C\min(1,\delta_J/\rho_x)$, and

$$\|h_x\|_2^2 \le C2^{J(d+4)}\min(1, 2^{-J}/\rho_x) \le C2^{J(d+3)}|x-x_J|^{-1}.$$

Since $\int_D |x-x_J|^{-1}dx \le C$ for $d \ge 2$, Tonelli's theorem and (4) give $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$. □

8.8 The remainder and the bad event

Lemma 8.13. *On $\mathcal{A}$, $\|T_{\hat V}^{-1}\operatorname{div}(Z\nabla\xi)\|_\infty \le CJ^4 2^{-J(2\alpha-1)}$. This is $o(2^{-J\alpha})$ because $\alpha > 1$.*

Proof. We use (5), (7), $(2-\alpha)_+ < 1$ and $(3-\alpha)_+ \le 2$. First,

$$\|G_\xi\|_\infty \le \|\bar f\|_\infty\|\Delta\xi\|_\infty + \|\nabla\bar f\|_\infty\|\nabla\xi\|_\infty \le CJ2^{-J(\alpha-1/2)} + CJ^2 2^{J(2-\alpha)_+ - J(\alpha+1/2)} \le CJ^2 2^{-J(\alpha-1/2)}.$$

Next, $\|\nabla G_\xi\|_\infty \le C(\|D^3\xi\|_\infty + \|\nabla\bar f\|_\infty\|D^2\xi\|_\infty + \|D^2\bar f\|_\infty\|\nabla\xi\|_\infty) \le CJ^3 2^{-J(\alpha-3/2)}$. Here each cross term has exponent $\le -(\alpha-3/2)$ because $\alpha \ge 1$.

By Lemma 8.7(b) and Lemma 8.9(d), with $\|D\operatorname{div} V_J\|_\infty \le d\|D^2 V_J\|_\infty \le CJ2^{J(2-\alpha)_+}$ from Lemma 8.4(3), we get $\|Z\|_\infty \le CJ^2 2^{-J(\alpha-1/2)}$ and $\|\nabla Z\|_\infty \le C(\|\nabla G_\xi\|_\infty + \|G_\xi\|_\infty J2^{J(2-\alpha)_+}) \le CJ^3 2^{-J(\alpha-3/2)}$. Hence

$$\|\operatorname{div}(Z\nabla\xi)\|_\infty \le \|\nabla Z\|_\infty\|\nabla\xi\|_\infty + \|Z\|_\infty\|\Delta\xi\|_\infty \le CJ^4 2^{-J(2\alpha-1)} + CJ^3 2^{-J(2\alpha-1)} \le CJ^4 2^{-J(2\alpha-1)}.$$

Lemma 8.7(b) for $X = \hat V$ concludes the proof. □

Lemma 8.14. *$P(\mathcal{A}^c) \le Ce^{-cJ^2}$, uniformly in f.*

Proof. Each component $\partial^\iota \xi(x) = n^{-1/2}\mathbb{W}(\partial_x^\iota k_J(x,\cdot))$, $|\iota| = m$, is centred Gaussian. By (K3) its standard deviation is $\le C\bar\sigma_m$, where $\bar\sigma_m := n^{-1/2}2^{J(d/2+m)}$, and its canonical metric is $\le C\bar\sigma_m 2^J|x-x'|$. The covering numbers of D in this metric are therefore $\le (C\bar\sigma_m 2^J/\eta)^d$, and Dudley's bound gives $\mathbb{E}\sup_D|\partial^\iota\xi| \le C\bar\sigma_m\sqrt{J} \le J\bar\sigma_m/2$ for large J. The Borell–TIS inequality at deviation $J\bar\sigma_m/2$ gives $P(\sup_D|\partial^\iota\xi| > J\bar\sigma_m) \le 2e^{-cJ^2}$. A union bound over the finitely many ι with $1 \le |\iota| \le 3$ concludes the proof. □

8.9 Assembly

Proof of Theorem 4.1(ii). Let $r \le r_0$ (the minimum of the values required in Lemmas 8.1 and 8.4) and $J \ge J_0$. On $\mathcal{A}$ we have $\mathcal{A} \subset \mathcal{B}$, so (8) applies. Clipping only decreases $|\tilde f - f|$, so Proposition 8.11 and Lemma 8.13 give

$$\|\tilde f - f\|_2^2 \le 3|D|\big(CJ^2 2^{-J\alpha}\big)^2 + 3\|Z\|_2^2 + 3|D|\big(CJ^4 2^{-J(2\alpha-1)}\big)^2.$$

Off $\mathcal{A}$, $|\tilde f - f| \le 2r$ pointwise, so $\|\tilde f - f\|_2^2 \le 4r^2|D|$. Taking expectations, using Proposition 8.12, Lemma 8.14 and (4),

$$\mathbb{E}_f\|\tilde f - f\|_2^2 \le CJ^4 2^{-2J\alpha} + C2^{-2J\alpha} + CJ^8 2^{-2J(2\alpha-1)} + Ce^{-cJ^2} \le CJ^4 2^{-2J\alpha} \le C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)},$$

uniformly over $\mathcal{F}(\alpha, r)$. The finitely many n with $J < J_0$ are covered by the trivial bound $4r^2|D|$. For non-integer α, Lemma 8.3 and (C3) produce no logarithm. Then the bias is $\le C2^{-J\alpha}$, and the remaining powers of J multiply terms of strictly smaller order. So the bound is $Cn^{-2\alpha/(2\alpha+d+3)}$. □

Theorem 4.1 follows from Sections 7 and 8.

> **Reviewer 2**
>
> Imprecise references.

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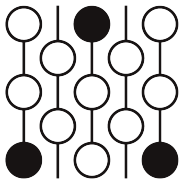
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Review of Problem 2 – Siege

First Proof Third Batch
Reviewer 3
October 8–9, 2026



1 Review

The approach presented in this manuscript is based on a plug-in kernel estimator for the solution u_f of the underlying PDE, combined with the corresponding inversion formula to reconstruct the unknown conductivity f, see Definition 8.2. The basic approach is comparable to the human-based solution but, notably, does not require an additional pre-estimation step. The risk analysis combines properties of the Kernel estimator with analytical properties of the transport inverse operator, see section 8.4. Compared to the human solution, the analysis proceeds directly in terms of the L^2 risk rather than the pointwise risk and employs a different error decomposition. The resulting upper bound is complemented by a lower bound established through a classical application of Assouad's lemma, with suitable perturbations constructed directly at the level of the conductivity f, rather than the solution u_f. The upper and lower bounds match up to logarithmic factors, establishing minimax optimality up to these terms. The heuristic discussion in Section 5 provides a convincing explanation of the convergence rate and its underlying mathematical structure. From a methodological perspective, the use of the Darcy inversion formula is particularly noteworthy, as this approach is itself novel in the field. Apart from differences in the error decomposition and the treatment of the L^2 risk, the solution deviates from the human-based approach mainly in minor aspects of the estimator construction and the proof of the lower bound. The overall estimation and proof strategy appear mathematically reliable. However, a complete verification of the proofs is considerably complicated by the notation and the organization of the arguments. While the exposition is generally comprehensible with substantial effort, its structure lacks clarity, with closely related statements distributed across several lemmas. Although the general strategy is outlined in Section 5, several essential proof ideas are insufficiently motivated or explained. In some places, such as Section 8.4 and Definition 8.6, the notation is unnecessarily elaborate, whereas other arguments are presented rather briefly, for instance in the proofs of Lemma 8.5 and Lemma 8.7. Furthermore, despite a few (accurate) references scattered throughout the manuscript, a complete literature review that provides a comprehensive overview of existing results and comparable approaches is necessary to meet scientific standards. This is not provided in the present manuscript.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors

- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

3 The problem

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, be the closed Euclidean unit ball. For $f \in C^\alpha(D)$ with Hölder–Zygmund smoothness $\alpha > 1$ (and f bounded away from 0), let $u = u_f$ be the unique solution of Darcy's problem

$$\operatorname{div}(f\nabla u) = 1 \ \text{ on } D, \qquad u|_{\partial D} = 0.$$

We observe

$$Y_n = u_f + \tfrac{1}{\sqrt{n}}\,\mathbb{W},$$

where $\mathbb{W}$ is a Gaussian white noise on $L^2(D)$. For $r \in (0,1)$ let

$$\mathcal{F}(\alpha, r) = \{h \in C^\alpha(D) : \ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r\},$$

where $\mathbf{1}$ is the constant function 1. For small but fixed $r \in (0,1)$, the question is: what is the minimax rate, as $n \to \infty$ and up to logarithmic factors, of

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\big[\|\hat{f}_n - f_0\|^2_{L^2(D)}\big] \ ?$$

4 Result

Theorem 4.1. *Fix $\alpha > 1$ and $d \geq 2$.*

(i) ***(Lower bound.)*** *For every $r \in (0,1)$ there are $c > 0$ and n_0 such that for $n \geq n_0$*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\hat{f} - f\|^2_{L^2(D)} \ \geq \ c\, n^{-2\alpha/(2\alpha+d+3)}.$$

(ii) ***(Upper bound.)*** *There is $r_0 = r_0(d,\alpha) > 0$ such that for every $r \in (0, r_0]$ an explicit estimator $\tilde{f}$ (Definition 8.2) satisfies, for $n \geq n_0$,*

$$\sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\tilde{f} - f\|^2_{L^2(D)} \ \leq \ C(\log n)^4\, n^{-2\alpha/(2\alpha+d+3)}.$$

If α is not an integer, the factor $(\log n)^4$ can be dropped.

The constants r_0, c, C, n_0 depend only on d, α, r and on fixed auxiliary profiles and kernels chosen below.

Corollary 4.2 (Answer)**.** *For every fixed $r \in (0, r_0]$, up to logarithmic factors,*

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_n - f_0\|^2_{L^2(D)} \ \asymp \ n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

Everything in Sections 7–8 is argued here; the only external inputs are the standard facts collected, with references, in Section 6.

5 Overview of the proof

Where $d+3$ comes from. Near $f = \mathbf{1}$ the solution is close to $u_1 := (|x|^2-1)/(2d)$, so $\nabla u_f \approx x/d$, a radial field with a single source at the origin. The linearised forward map $h \mapsto -L_f^{-1}\operatorname{div}(h\nabla u_f)$, with $L_f = \operatorname{div}(f\nabla\cdot)$ (Dirichlet), therefore smooths by order 2 for perturbations oscillating across the rays (in the $d-1$ angular directions), but only by order 1 along the rays, because $\operatorname{div}(h\,x/d)$ is a first-order operator along the rays. Heuristically, the symbol of the linearisation at $f = \mathbf{1}$ is $\asymp (1+|\xi_{\mathrm{rad}}|)/|\xi|^2$, and the exponent $d+3$ lies between the isotropic values $d+4$ and $d+2$. This heuristic only motivates the construction; it is not used in the proofs.

Lower bound. We perturb $\mathbf{1}$ by $M \asymp 2^{j(d-1)}$ disjoint bumps that oscillate at frequency 2^j in the angular directions only. Each has the form $2^{-j(\alpha+2)}b(|x|)\Delta\Phi_k$ with Φ_k homogeneous of degree 0; because of this form, the radial transport term $\operatorname{div}(h\,x/d)$ is again a Laplacian of a function of size $2^{-j(\alpha+2)}$. A duality argument then shows that the forward map shrinks each bump by 2^{-2j} in L^2. Balancing the Kullback–Leibler divergence with Assouad's lemma gives $2^j \asymp n^{1/(2\alpha+d+3)}$ and risk $2^{-2j\alpha}$.

Upper bound. We smooth the data at resolution 2^{-J}, $2^J \asymp n^{1/(2\alpha+d+3)}$, and estimate ∇u by $\hat V = \nabla K_J Y$. For r small, $\hat V$ is C^1-close to x/d, and the transport equation $\operatorname{div}(g\hat V) = 1$ has an explicit solution $T_{\hat V}^{-1}\mathbf{1}$, obtained by integrating along the backward flow of $\hat V$, which runs into a single sink. We linearise around the deterministic centring function $\bar f = T_{V_J}^{-1}\mathbf{1}$, where $V_J = \nabla K_J u$. This gives an exact decomposition: a bias $\bar f - f = O(J^2 2^{-J\alpha})$, a linear Gaussian term Z and a quadratic remainder. The variance of Z is the key point. Z integrates $\Delta\xi$ along rays, where ξ is the smoothed noise. Now $\Delta\xi$ has pointwise variance $n^{-1}2^{J(d+4)}$ and correlation length 2^{-J}, and integrating it along a ray of length $O(1)$ gains one factor 2^{-J}. Hence $\mathbb{E}\|Z\|_2^2 \lesssim n^{-1}2^{J(d+3)}$: the noise is smoothed by one order along the flow, and this produces $d+3$. The quadratic remainder is negligible because $\alpha > 1$.

6 Conventions and standard facts

C^s denotes the Hölder–Zygmund space $B^s_{\infty\infty}$; on D it consists of restrictions, with the quotient norm. Constants c, C may change from line to line. They never depend on j, J, n or on $f \in \mathcal{F}(\alpha, r)$. D° denotes the open unit ball. We write $u_1 := (|x|^2-1)/(2d)$, the solution for $f = \mathbf{1}$, so that $\nabla u_1 = x/d$. For $g \geq 1-r$ write $L_g = \operatorname{div}(g\nabla\cdot)$ with Dirichlet boundary conditions. Expectation is $\mathbb{E}$, and P_f is the law of Y_n. In the white-noise model, by the Cameron–Martin theorem (see e.g. [6]),

$$\mathrm{KL}(P_f, P_g) = \tfrac{n}{2}\|u_f - u_g\|_{L^2}^2.$$

We use the following known results without proof.

> **Reviewer 3**
>
> Here and in the following, notation for multidimensional derivatives is used incorrectly, since multi-index notation has not been introduced and/or is being mixed with one-dimensional derivatives.

(E1) *Global H^2 regularity* [5, Theorem 8.12], combined with the energy estimate. Let $g \in C^{0,1}(D)$ with $g \geq 1-r$. The solution $w \in H_0^1(D)$ of $\operatorname{div}(g\nabla w) = F \in L^2$ satisfies $\|w\|_{H^2} \leq C\|F\|_2$, where C depends only on d, $1-r$ and an upper bound for $\|g\|_{C^{0,1}}$.

(E2) *$W^{2,p}$ theory, $1 < p < \infty$* [5, Theorem 9.15, Lemma 9.17]. Let $g \in C^1(D)$ with $g \geq 1 - r$. The Dirichlet problem $g\Delta w + \nabla g \cdot \nabla w = F \in L^p$, $w \in W^{2,p} \cap W^{1,p}_0$, has a unique solution, and $\|w\|_{W^{2,p}} \leq C\|F\|_p$. The constant is uniform over $\mathcal{F}(\alpha, r)$, which is bounded in C^α and hence precompact in C^1.

(GN) *Gagliardo–Nirenberg inequality on the ball* [4, 8]. For $p > d$ and $t = (1 + d/p)/2$, $\|\nabla w\|_\infty \leq C\|w\|^t_{W^{2,p}}\|w\|^{1-t}_p + C\|w\|_p$.

(As) *Assouad's lemma* [1]; see also [12, Ch. 2]. Let $\{P_\theta : \theta \in \{0,1\}^M\}$ be probability measures with $\mathrm{TV}(P_\theta, P_{\theta'}) \leq \tau < 1$ whenever θ, θ' differ in exactly one coordinate. Then for every estimator $\hat\theta$ with values in $\{0,1\}^M$, $\max_\theta \mathbb{E}_\theta \sum_k \mathbf{1}\{\hat\theta_k \neq \theta_k\} \geq \frac{M}{2}(1-\tau)$. We also use Pinsker's inequality $\mathrm{TV} \leq \sqrt{\mathrm{KL}/2}$ [12, Ch. 2].

(C1) *Seeley-type extension* [9]; see also [10]. There is a linear operator E with $Eg = g$ on D that is bounded $C^s(D) \to C^s(\mathbb{R}^d)$ for $0 < s \leq S_0$, where we fix $S_0 > \alpha + 2$. It has the form $Eg(t) = \chi(|t|)\sum_i a_i\, g(R_i t)$ for $|t| > 1$, where $R_i t = (1 - \lambda_i(|t| - 1))\, t/|t|$, finitely many $\lambda_i > 0$, and χ is a smooth cutoff vanishing for $|t| \geq 1 + \eta$, with $\lambda_i \eta < 1$.

(C2) *Kernel approximation* (see e.g. [6]). Let $k \in C^\infty_c(\mathbb{R}^d)$ with $\int k = 1$ and vanishing moments of orders $1, \ldots, N_0$, where we fix an integer $N_0 \geq 2\alpha + 2$. Put $k_J = 2^{Jd}k(2^J\cdot)$. Then $\|D^m(k_J * g - g)\|_\infty \leq C2^{-J(s-m)}\|g\|_{C^s}$ for integers $0 \leq m < s$ and $0 < s \leq N_0$.

(LP) *Littlewood–Paley theory* [10, 6]. For a fixed Littlewood–Paley decomposition $g = \sum_{j\geq 0}\Delta_j g$ we have $\|g\|_{C^s} \asymp \sup_j 2^{js}\|\Delta_j g\|_\infty$ for $s > 0$. Bernstein's inequality gives $\|D^m\Delta_j g\|_\infty \leq C2^{jm}\|\Delta_j g\|_\infty$. Moreover $C^s \subset C^{1,\beta_0}$ for $s > 1$, with $\beta_0 = \min(s - 1, 1/2)$.

(C3) *Taylor formula in Zygmund spaces; algebra property* (standard; see [10]). Let $\phi \in C^s$, $s > 0$, and $|h| \leq 1$. Then $\Delta_h\phi(x) := \phi(x + h) - \phi(x) = P_\phi(x, h) + r_\phi(x, h)$, where $P_\phi(x, h) = \sum_{1\leq|\iota|<s} \partial^\iota\phi(x)h^\iota/\iota!$ and $|r_\phi(x, h)| \leq C\|\phi\|_{C^s}|h|^s\Lambda(h)$ with $\Lambda(h) = 1+\log(1/|h|)$; the logarithm is needed only when s is an integer. Also $C^s \cdot C^s \subset C^s$, with $\|\phi\psi\|_{C^s} \leq C\|\phi\|_{C^s}\|\psi\|_{C^s}$.

(S) *Schauder estimate in Zygmund spaces* (standard). The Dirichlet Laplacian satisfies: $\Delta_D^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ is bounded for $\alpha - 1 > 0$. For non-integer α this is classical Schauder theory [5, Ch. 6].

(ODE) *C^k dependence of flows on initial data*, with the variational equations [7, Ch. V].

(G) *Gaussian facts* [6]. Let $y \mapsto \varphi_y \in L^2(D)$ be C^{m+1} on an open set, with locally Lipschitz derivatives. Then $\mathbb{W}(\varphi_y)$ has a C^m version with $\partial^\iota_y\mathbb{W}(\varphi_y) = \mathbb{W}(\partial^\iota_y\varphi_y)$ for $|\iota| \leq m$. Bochner integrals in L^2 commute with $\mathbb{W}$. We also use Dudley's entropy bound [3] and the Borell–Tsirelson–Ibragimov–Sudakov (Borell–TIS) inequality [2, 11].

7 The lower bound

Throughout this section $\alpha > 1$, $d \geq 2$ and $r \in (0, 1)$ are fixed, and $j \geq j_0$ is large.

7.1 The hypercube

Directions and bumps. Choose $\omega_1, \ldots, \omega_M \in S^{d-1}$ with pairwise geodesic distance $\geq 2 \cdot 2^{-j}$ and $M \geq c_d 2^{j(d-1)}$ (a maximal separated set). Fix $\varphi_0 \in C^\infty_c([0, 1/2))$ that is constant near 0 and not identically 0. Let $\Phi_k(\omega) = \varphi_0(2^j d_S(\omega, \omega_k))$, where d_S is the geodesic distance. The Φ_k are

smooth, are rotations of one another, and have pairwise disjoint supports, which are caps of radius $2^{-j}/2$.

Extension and scaling. Extend each Φ_k to $\mathbb{R}^d \setminus \{0\}$ as a function homogeneous of degree 0. Then

$$\Delta\Phi_k(x) = |x|^{-2}(\Delta_S\Phi_k)(x/|x|),$$

where Δ_S is the Laplace–Beltrami operator. Put $\chi_k := 2^{-2j}\Delta_S\Phi_k$. In rescaled normal coordinates this is the flat Laplacian of a fixed nonzero compactly supported bump, up to $O(2^{-j})$ metric corrections. Hence $\|\partial^m\chi_k\|_\infty \le C_m 2^{jm}$, and for $j \ge j_0$

$$c2^{-j(d-1)} \le \|\chi_k\|^2_{L^2(S^{d-1})} \le C2^{-j(d-1)}.$$

The lower bound holds because the Laplacian of a nonzero compactly supported function is nonzero, and the value does not depend on k by rotation invariance.

The perturbations. Fix $b \in C_c^\infty((1/4,3/4))$, not identically 0. Define

$$h_k(x) := \kappa 2^{-j(\alpha+2)} b(|x|)\,\Delta\Phi_k(x) = \kappa 2^{-j\alpha} b(|x|)|x|^{-2}\chi_k(x/|x|), \qquad f_\theta := \mathbf{1} + \sum_{k=1}^{M} \theta_k h_k, \quad \theta \in \{0,1\}^M.$$

Lemma 7.1. *There is $\kappa = \kappa(r) > 0$ such that $f_\theta \in \mathcal{F}(\alpha, r)$ for all θ and all j. The supports S_k of the h_k are pairwise disjoint, and*

$$\|h_k\|_2^2 \asymp 2^{-j(2\alpha+d-1)}, \qquad \|\Phi_k\|_{L^2(\{1/4<|x|<3/4\})} \asymp 2^{-j(d-1)/2}, \qquad \Big\|\sum_k \theta_k\Phi_k\Big\|_\infty \le \|\varphi_0\|_\infty.$$

Proof. Let $G_\theta = \sum_k \theta_k\chi_k$. Each point of the sphere has a neighbourhood meeting at most one support, so $\|D^m G_\theta\|_\infty \le C2^{jm}$ and $\|G_\theta\|_\infty \le C$. Take an integer $m > \alpha$. Then $|\Delta_h^m(2^{-j\alpha}G_\theta)| \le C2^{-j\alpha}\min((2^j|h|)^m, 1) \le C|h|^\alpha$. Hence $\|2^{-j\alpha}G_\theta\|_{C^\alpha(S^{d-1})} \le C$ uniformly in θ and j. Composing with $x \mapsto x/|x|$ and multiplying by the smooth function $b(|x|)|x|^{-2}$, supported in the annulus, keeps this bounded. Choosing κ small gives $\|f_\theta - \mathbf{1}\|_{C^\alpha(D)} \le r$. Disjointness of the supports is clear. In polar coordinates, $\|h_k\|_2^2 = \kappa^2 2^{-2j\alpha}\int b(\rho)^2\rho^{d-5}d\rho\,\|\chi_k\|^2_{L^2(S^{d-1})} \asymp 2^{-j(2\alpha+d-1)}$. The remaining two statements are immediate from the construction. □

By disjointness, $\|f_\theta - f_{\theta'}\|_2^2 = \sum_{k:\theta_k\neq\theta'_k}\|h_k\|_2^2$.

7.2 The key estimate

Proposition 7.2. *If $\theta, \theta' \in \{0,1\}^M$ differ only in coordinate k, then $\|u_{f_\theta} - u_{f_{\theta'}}\|_2 \le C2^{-2j}\|h_k\|_2$. Consequently $\|u_{f_\theta} - u_{f_{\theta'}}\|_2^2 \le C2^{-j(2\alpha+d+3)}$.*

Proof. Exact identity. Let $f = f_{\theta'}$, $g = f_\theta = f + h$ with $h = \pm h_k$, and $V = \nabla u_f$. Subtracting the two equations gives $\operatorname{div}(g\nabla(u_g - u_f)) = 1 - \operatorname{div}(g\nabla u_f) = -\operatorname{div}(hV)$. Hence

$$u_g - u_f = -L_g^{-1}\operatorname{div}(hV).$$

(i) Duality. L_g is self-adjoint on H_0^1, so $\|L_g^{-1}F\|_2 = \sup_{\|\phi\|_2\le 1}\langle F, \psi\rangle$ with $\psi := L_g^{-1}\phi$. Since $g \ge 1 - r$ and g is Lipschitz (as $\alpha > 1$), (E1) gives $\|\psi\|_{H^2} \le C$. It therefore suffices to show

$$|\langle \operatorname{div}(h_k V), \psi\rangle| \le C2^{-2j}\|h_k\|_2\|\psi\|_{H^2} \qquad \text{for } \psi \in H^2 \cap H_0^1. \tag{1}$$

Split $V = x/d + \nabla v$ with $v := u_f - u_1$.

(ii) Radial part. For $G = b(|x|)\Delta\Phi_k$ we claim

$$\operatorname{div}(G\,x/d) = b_1(|x|)\Delta\Phi_k, \qquad b_1(\rho) := \frac{\rho b'(\rho) + (d-2)b(\rho)}{d} \in C_c^\infty((1/4, 3/4)). \tag{2}$$

Indeed, $\Psi := \Delta\Phi_k$ is homogeneous of degree -2, so Euler's relation gives $x \cdot \nabla G = \rho b'\Psi - 2b\Psi$. Since $\operatorname{div}(G\,x) = x \cdot \nabla G + dG$, dividing by d gives (2). The function $b_1\psi$ has compact support in the annulus, where Φ_k is smooth. Integrating by parts twice and using Lemma 7.1,

$$|\langle \operatorname{div}(h_k\,x/d), \psi\rangle| = \kappa 2^{-j(\alpha+2)}\Big|\int \Phi_k\,\Delta(b_1\psi)\Big| \le C2^{-j(\alpha+2)}2^{-j(d-1)/2}\|\psi\|_{H^2} \le C2^{-2j}\|h_k\|_2\|\psi\|_{H^2}.$$

(iii) Drift part. Since h_k has compact support, $|\langle \operatorname{div}(h_k\nabla v), \psi\rangle| = |\int h_k\nabla v\cdot\nabla\psi| \le \|h_k\|_2\|\nabla v\|_\infty\|\psi\|_{H^1}$. It therefore suffices to show $\|\nabla v\|_\infty \le C2^{-j(\alpha+1-d/p)}$ for every $p > d$, and then to take $p > d$ with $d/p \le \alpha - 1$.

Equation for v. We have $\operatorname{div}(f\nabla v) = 1 - \operatorname{div}(f\,x/d) = -\operatorname{div}((f-1)x/d) =: -q$. By (2) applied termwise, $q = \kappa 2^{-j(\alpha+2)}b_1(|x|)\Delta\Phi_{\theta'}$ with $\Phi_{\theta'} = \sum_l \theta'_l\Phi_l$.

$W^{2,p}$ bound. Since $q = \kappa 2^{-j\alpha}b_1(|x|)|x|^{-2}\sum_l \theta'_l\chi_l(x/|x|)$ and the χ_l are bounded with disjoint supports, $\|q\|_p \le C2^{-j\alpha}$. Hence $\|v\|_{W^{2,p}} \le C2^{-j\alpha}$ by (E2).

L^p bound. Take ϕ with $\|\phi\|_{p'} \le 1$ and let $\psi = L_f^{-1}\phi \in W^{2,p'} \cap W_0^{1,p'}$, so that $\|\psi\|_{W^{2,p'}} \le C$ by (E2). By Green's identity, and integrating by parts twice as in (ii),

$$\Big|\int v\phi\Big| = \Big|\int q\psi\Big| = \kappa 2^{-j(\alpha+2)}\Big|\int \Phi_{\theta'}\,\Delta(b_1\psi)\Big| \le C2^{-j(\alpha+2)}\|\varphi_0\|_\infty\|\psi\|_{W^{2,p'}} \le C2^{-j(\alpha+2)}.$$

So $\|v\|_p \le C2^{-j(\alpha+2)}$.

Interpolation. By (GN), $\|\nabla v\|_\infty \le C2^{-j\alpha t}2^{-j(\alpha+2)(1-t)} + C2^{-j(\alpha+2)} \le C2^{-j(\alpha+1-d/p)}$, since $2(1-t) = 1 - d/p$.

Combining (ii) and (iii) proves (1), hence the first claim. The second follows from $\|h_k\|_2^2 \le C2^{-j(2\alpha+d-1)}$. □

7.3 Conclusion via Assouad's lemma

Proof of Theorem 4.1(i). By Proposition 7.2, $\mathrm{KL}(P_{f_\theta}, P_{f_{\theta'}}) \le Cn2^{-j(2\alpha+d+3)}$ for neighbours θ, θ'. Choose $2^j \asymp n^{1/(2\alpha+d+3)}$, with a constant large enough that $\mathrm{KL} \le 1/8$ for all neighbours. Pinsker's inequality then gives $\mathrm{TV} \le 1/4$.

Given any estimator $\hat f$, let $\hat\theta_k$ minimise $\|(\hat f - 1 - th_k)\mathbf{1}_{S_k}\|_2$ over $t \in \{0, 1\}$. On S_k we have $f_\theta = 1 + \theta_k h_k$. If $\hat\theta_k \ne \theta_k$, the triangle inequality gives $\|h_k\|_2 \le \|(\hat f - 1 - \hat\theta_k h_k)\mathbf{1}_{S_k}\|_2 + \|(\hat f - 1 - \theta_k h_k)\mathbf{1}_{S_k}\|_2 \le 2\|(\hat f - f_\theta)\mathbf{1}_{S_k}\|_2$. Since the S_k are disjoint,

$$\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \sum_k \|h_k\|_2^2\,\mathbf{1}\{\hat\theta_k \ne \theta_k\}.$$

By (As) with $\tau = 1/4$ and Lemma 7.1,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\hat f - f\|_2^2 \ge \max_\theta \mathbb{E}_\theta\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \min_k \|h_k\|_2^2 \cdot \tfrac{M}{2}\cdot\tfrac34 \ge c2^{j(d-1)}2^{-j(2\alpha+d-1)} = c2^{-2j\alpha} \asymp n^{-2\alpha/(2\alpha+d+3)}.$$

The condition $\alpha > 1$ was used only in (iii), and to make $f \in C^1$ for (E1) and (E2). □

8 The upper bound

> **Reviewer 3**
>
> The presentation of the entire section is unnecessarily complicated and makes it difficult to understand the approach and verify its correctness. A reorganization that clearly presents the estimation approach, followed by a proof of its properties, is necessary.

Throughout, $\alpha > 1$ and $d \geq 2$ are fixed, $f \in \mathcal{F}(\alpha, r)$, and $r \leq r_0$, where r_0 is reduced finitely many times below.

8.1 A Schauder bound

Lemma 8.1. *There is $r_0 = r_0(d, \alpha)$ such that for $r \leq r_0$ and all $f \in \mathcal{F}(\alpha, r)$,*

$$\|u_f - u_1\|_{C^{\alpha+1}(D)} \leq Cr, \qquad \|u_f\|_{C^{\alpha+1}(D)} \leq C. \tag{3}$$

Proof. Write $L_f = \Delta + \operatorname{div}((f-1)\nabla\cdot)$. By the algebra property (C3), $\|\operatorname{div}((f-1)\nabla v)\|_{C^{\alpha-1}} \leq C_1\|f-1\|_{C^\alpha}\|v\|_{C^{\alpha+1}} \leq C_1 r\|v\|_{C^{\alpha+1}}$. Take r_0 with $C_1 r_0 \|\Delta_D^{-1}\| \leq 1/2$. Writing $L_f v = F$ as $(I + \Delta_D^{-1}\operatorname{div}((f-1)\nabla\cdot))v = \Delta_D^{-1}F$ on the closed subspace of $C^{\alpha+1}(D)$ with zero trace, a Neumann series using (S) shows that $L_f^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ (Dirichlet data) has norm $\leq 2\|\Delta_D^{-1}\|$. The solution so obtained is classical, hence equal to the weak solution. As in Proposition 7.2, $u_f - u_1 = -L_f^{-1}\operatorname{div}((f-1)x/d)$, and the right-hand side has $C^{\alpha-1}$ norm $\leq Cr$. □

> **Reviewer 3**
>
> To the best of my knowledge, the techniques described here are similar to those in Nickl, van de Geer, and Wang (2020). Even when presenting a self-contained proof, this paper should be cited here.

This is the first place where smallness of r enters.

8.2 Smoothing, noise and the estimator

Resolution and smoother. Let $J := \lfloor \log_2 n/(2\alpha + d + 3)\rfloor$. Then

$$n^{-1/2} \leq 2^{-J(\alpha+(d+3)/2)}, \qquad n^{-1}2^{J(d+3)} \leq 2^{-2J\alpha}, \qquad 2^{-2J\alpha} \leq 2^{2\alpha} n^{-2\alpha/(2\alpha+d+3)}, \qquad J \leq \log_2 n. \tag{4}$$

Put $K_J g := (k_J * Eg)|_D$, with k_J from (C2) and E from (C1). On data, $K_J Y(x) = \langle Y, k_J(x,\cdot)\rangle$, where $k_J(x,\cdot) := E^*[k_J(x-\cdot)] \in L^2(D)$; this is defined for all $x \in \mathbb{R}^d$. By the change of variables $z = R_i t$, E^* is the identity on D plus finitely many terms of the form $a_i(\chi\varphi)(R_i^{-1}z)|\det DR_i^{-1}(z)|$. These are supported on shells $\{1 - \lambda_i\eta < |z| < 1\}$ and have bounded Jacobians. This gives the following kernel bounds.

(K1) $|\partial_x^\iota k_J(x,z)| \leq C2^{J(d+|\iota|)}$ for $|\iota| \leq 4$.

(K2) $\partial_x^\iota k_J(x,z) = 0$ unless $|x - z| \leq C_0 2^{-J}$. Indeed, for the reflected terms let $t = R_i^{-1}z$, so $|t| > 1$. If $x \in D$ and $|x-t| \leq C2^{-J}$, then $|t|-1 \leq C2^{-J}$ and $|z-t| = (1+\lambda_i)(|t|-1) \leq C2^{-J}$.

(K3) $\|\partial_x^\iota k_J(x,\cdot)\|_2^2 \leq C2^{J(d+2|\iota|)}$, and $x \mapsto \partial_x^\iota k_J(x,\cdot)$ is Lipschitz into L^2 with constant $C2^{J(d/2+|\iota|+1)}$, for $|\iota| \leq 3$. This follows from (K1), (K2) and the fact that the support in z has volume $\leq C2^{-Jd}$.

> **Reviewer 3**
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> Even if this is not a formal error, it appears that the notation commonly used in the literature for kernel methods is being combined here with the standard notation from wavelet theory, which can lead to unnecessary confusion and should be avoided.

Notation. Let $U = Eu$, $F = Ef$, $V = \nabla u$ and $V^E = \nabla U$. By (C1) and (3), $\|F\|_{C^\alpha} + \|V^E\|_{C^\alpha} \le C$. Let

$$V_J := \nabla K_J u = (k_J * V^E)|_D, \qquad e := K_J u - u, \qquad \xi := n^{-1/2} K_J \mathbb{W}, \qquad \hat{V} := \nabla K_J Y = V_J + \nabla\xi.$$

V_J is the restriction of a C^∞ field on $\mathbb{R}^d$. By (G) and (K1)–(K3), ξ is a centred Gaussian field with C^3 sample paths on a neighbourhood of D (it is in fact smooth), and ξ does not depend on f.

Good event. Let

$$\mathcal{A} := \Big\{ \max_{|\iota|=m} \sup_D |\partial^\iota \xi| \le J n^{-1/2} 2^{J(d/2+m)},\ m = 1,2,3 \Big\}.$$

On $\mathcal{A}$, by (4) and after adjusting constants for operator norms,

$$\|\nabla\xi\|_\infty \le CJ2^{-J(\alpha+1/2)}, \qquad \|D^2\xi\|_\infty \le CJ2^{-J(\alpha-1/2)}, \qquad \|D^3\xi\|_\infty \le CJ2^{-J(\alpha-3/2)}. \tag{5}$$

Definition 8.2 (Estimator)**.** Let $\varepsilon_0 := 1/(8d)$, and define the observable event $\mathcal{B} := \{\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon_0\}$. On $\mathcal{B}$ set $\hat{f} := T_{\hat{V}}^{-1}\mathbf{1}$ (Definition 8.6); off $\mathcal{B}$ set $\hat{f} := \mathbf{1}$. Let $\tilde{f} := \min(\max(\hat{f}, 1-r), 1+r)$.

> **Reviewer 3**
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> Notation is used here that does not appear until later in the text.

Since $\|f - 1\|_\infty \le r$, clipping can only decrease $|\tilde{f} - f|$ pointwise. The estimator uses only Y, n, α, r and the fixed kernels. The event $\mathcal{A}$ itself is not observable: ξ does not depend on f, but $\xi = K_J Y - K_J u$ cannot be computed from the data because u is unknown. This is why the estimator is defined on $\mathcal{B}$; Lemma 8.4 shows $\mathcal{A} \subset \mathcal{B}$.

8.3 The smoothed fields

Lemma 8.3. *Let $g \in C^s(\mathbb{R}^d)$, $s > 0$, and let $m \ge 1$ be an integer. Then $\|D^m(k_J * g)\|_\infty$ is at most $C2^{J(m-s)}\|g\|_{C^s}$ if $m > s$, at most $C(J+1)\|g\|_{C^s}$ if $m = s$, and at most $C\|g\|_{C^s}$ if $m < s$.*

Proof. By (LP), $D^m(k_J * g) = \sum_j (D^m k_J) * \Delta_j g$, and the series converges uniformly because each term is $\le \|D^m k_J\|_1 \|\Delta_j g\|_\infty \le C2^{Jm}2^{-js}\|g\|_{C^s}$. For $j \le J$, write the term as $k_J * D^m\Delta_j g$; by Bernstein it is $\le C2^{j(m-s)}\|g\|_{C^s}$. For $j > J$, the first bound gives $C2^{Jm-js}\|g\|_{C^s}$, and summing over $j > J$ gives $C2^{J(m-s)}\|g\|_{C^s}$. Summing the bound $C2^{j(m-s)}$ over $j \le J$ gives the three cases. □

Lemma 8.4. *Let $\varepsilon \le \varepsilon_0$. There are r_0 and J_0 such that for $r \le r_0$ and $J \ge J_0$:*

(1) $\|V_J - x/d\|_{C^1(D)} \le \varepsilon/2$;

(2) on $\mathcal{A}$, $\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon$; in particular (with $\varepsilon = \varepsilon_0$) $\mathcal{A} \subset \mathcal{B}$;

(3) $\|D^m V_J\|_\infty \le CJ2^{J(m-\alpha)_+}$ for $m = 2, 3$, with no factor J unless $\alpha = m$.

Proof. Item (3) is Lemma 8.3 with $g = V^E$ and $s = \alpha$. For (1), on D we have $V_J - x/d = (k_J * V^E - V^E) + \nabla(u - u_1)$. By (LP), $D^i V^E$ is β_0-Hölder for $i = 0, 1$, with $\beta_0 = \min(\alpha - 1, 1/2)$. Hence $|D^i(k_J * V^E - V^E)| \le C\int |k_J(y)||y|^{\beta_0} dy \le C2^{-J\beta_0}$. By (3), $\|\nabla(u - u_1)\|_{C^1} \le Cr$. Take r_0 and J_0 such that $C(2^{-J_0\beta_0} + r_0) \le \varepsilon/2$. For (2), on $\mathcal{A}$, (5) gives $\|\nabla\xi\|_{C^1} \le CJ2^{-J(\alpha-1/2)} \le \varepsilon/2$ for $J \ge J_0$. □

8.4 Geometry of the flow and the transport inverse

Setting. Let X be the restriction to D of a C^k field ($k \ge 1$) on a neighbourhood of D, with

$$\|X - x/d\|_{C^1(D)} := \max\big(\sup_D |X - x/d|,\ \sup_D \|DX - I/d\|\big) \le \varepsilon \le 1/(8d).$$

Put $\nu := 1/d - \varepsilon \ge 7/(8d)$. Let Φ_t be the flow of X, let $\gamma(s) = \gamma_x(s) = \Phi_{-s}x$ be the backward ray from x, and let

$$w(s,x) := \exp\Big(-\int_0^s \operatorname{div} X(\Phi_{-\sigma}x)\,d\sigma\Big).$$

Lemma 8.5. *In this setting the following hold.*

(1) X has exactly one zero $x_X \in D$, and $|x_X| \le d\varepsilon$.

(2) With $\rho(y) = |y - x_X|$: $(y - x_X)\cdot X(y) \ge (1/d - \varepsilon)\rho^2$ and $|X(y)| \le (1/d + \varepsilon)\rho$ on D.

(3) For $x \in D$, $\Phi_{-s}x$ exists for all $s \ge 0$ and stays in D. It lies in D° for $s > 0$, and D° is invariant.

(4) $\varrho(s) := \rho(\gamma(s))$ satisfies $\varrho(0)e^{-(1/d+\varepsilon)s} \le \varrho(s) \le \varrho(0)e^{-(1/d-\varepsilon)s}$. If $\varrho(0) > 0$, then $-(1/d + \varepsilon)\varrho \le \varrho' \le -\nu\varrho$.

(5) The propagator $P(s,\sigma)$ of $y' = -DX(\gamma)y$ satisfies $\|P(s,\sigma)\| \le e^{-\nu(s-\sigma)}$. In particular $\|D\Phi_{-s}\| \le e^{-\nu s}$.

(6) $1 - d\varepsilon \le \operatorname{div} X \le 1 + d\varepsilon$, so $e^{-(1+d\varepsilon)s} \le w \le e^{-(1-d\varepsilon)s} \le e^{-7s/8}$.

Proof. (1) The map $x \mapsto x - dX(x) = -d(X - x/d)$ sends D into $B(0, d\varepsilon) \subset D$ and is a $d\varepsilon$-contraction on the convex set D. Its unique fixed point is the unique zero of X. (2) $X(y) = Q(y)(y - x_X)$ with $Q = \int_0^1 DX(x_X + t(y - x_X))\,dt$ and $\|Q - I/d\| \le \varepsilon$; the segment lies in D by convexity. (3) For $|z| = 1$, $z \cdot X(z) \ge 1/d - \varepsilon > 0$, so $\frac{d}{ds}|\gamma|^2 = -2\gamma\cdot X(\gamma) < 0$ whenever $|\gamma| = 1$. A first-exit-time argument gives invariance of D and of D°, and shows that boundary points enter D° immediately. Global existence holds because D is compact. (4) $(\varrho^2)' = -2(\gamma - x_X)\cdot X(\gamma)$; apply (2) and Gronwall's inequality. (5) $v \cdot DX\, v \ge \nu|v|^2$ on D, so $(|y|^2)' \le -2\nu|y|^2$. Also $D\Phi_{-s} = P(s,0)$. (6) $\operatorname{div} X = 1 + \operatorname{tr}(DX - I/d)$ and $|\operatorname{tr} A| \le d\|A\|$. □

Definition 8.6. For $g \in C^1(D)$ put $T_X g := \operatorname{div}(gX)$. For $H \in C(D)$ put

$$T_X^{-1}H(x) := \int_0^\infty H(\Phi_{-s}x)\,w(s,x)\,ds.$$

By Lemma 8.5(6) the integral converges absolutely and uniformly, and $T_X^{-1}H \in C(D)$.

Since $w(s, \Phi_{-t}x) = w(s+t,x)/w(t,x)$, we have the semigroup identity

$$w(t,x)\,T_X^{-1}H(\Phi_{-t}x) = \int_t^\infty H(\Phi_{-s}x)\,w(s,x)\,ds. \tag{6}$$

Lemma 8.7. *In the setting of Lemma 8.5:*

(a) (Uniqueness.) *If $g \in C^1(D)$ and $\operatorname{div}(gX) = H$ on D° with $H \in C(D)$, then $g = T_X^{-1}H$.*

(a′) (Right inverse.) *If $H \in C(D)$ and $g := T_X^{-1}H \in C^1(D)$, then $\operatorname{div}(gX) = H$ on D.*

(b) $\|T_X^{-1}H\|_\infty \le \frac{8}{7}\|H\|_\infty \le 2\|H\|_\infty$.

(c) (Boundary layer.) *If $H = 0$ on $\{|x| \le 1-\delta\}$, then $\|T_X^{-1}H\|_\infty \le 6d\delta\|H\|_\infty$.*

Proof. (a) Fix $x \in D$. For $s > 0$, $\gamma(s) \in D^\circ$, and $\psi(s) = g(\gamma(s))w(s,x)$ satisfies $\psi' = -(X\cdot\nabla g + g\,\mathrm{div}\,X)(\gamma)\,w = -H(\gamma)\,w$. Integrate over $[\delta, S]$, then let $\delta \to 0$ (by continuity) and $S \to \infty$ (using $|\psi(S)| \le \|g\|_\infty e^{-7S/8}$). This gives $g(x) = \int_0^\infty H(\gamma(s))w(s,x)\,ds$.

(a′) Fix $x \in D^\circ$ and differentiate (6) at $t = 0^+$. The right side gives $-H(x)$. Since $w(0,x) = 1$ and $\partial_t w(0,x) = -\,\mathrm{div}\,X(x)$, the left side gives $-\nabla g(x)\cdot X(x) - g(x)\,\mathrm{div}\,X(x)$. Extend to D by continuity.

(b) Use $w \le e^{-7s/8}$.

(c) The case $\delta > 1/2$ follows from (b), since then $6d\delta > 2$. Let $\delta \le 1/2$ and $\phi = |\gamma|^2$. Where $|\gamma| \ge 1/2$, $\gamma\cdot X(\gamma) \ge |\gamma|(|\gamma|/d - \varepsilon) \ge 3/(16d)$, so $\phi' \le -3/(8d)$. Hence ϕ drops below $(1-\delta)^2$ within time $\le (1-(1-\delta)^2)\cdot 8d/3 \le 6d\delta$, and by the same sign argument it never returns. Since $H(\gamma(s)) = 0$ afterwards and $w \le 1$, the claim follows. □

Lemma 8.8 (Flow derivatives, uniform in s)**.** *Let $M = D\Phi_{-s}$, $N = D^2\Phi_{-s}$ and $\ell = \log w$, all differentiated in x.*

(1) If $X \in C^2$: $\|N(s)\| \le \|D^2X\|_\infty e^{-\nu s}/\nu$ and $|D_x\ell| \le \|D\,\mathrm{div}\,X\|_\infty/\nu$.

(2) If $X \in C^3$: $\|D_x^2\ell\| \le \|D^2\,\mathrm{div}\,X\|_\infty/(2\nu) + \|D\,\mathrm{div}\,X\|_\infty\|D^2X\|_\infty/\nu^2$.

Proof. N solves $N' = -DX(\gamma)N - D^2X(\gamma)[M\cdot, M\cdot]$ with $N(0) = 0$. By Duhamel's formula and Lemma 8.5(5), $\|N(s)\| \le \int_0^s e^{-\nu(s-\sigma)}\|D^2X\|_\infty e^{-2\nu\sigma}d\sigma \le \|D^2X\|_\infty e^{-\nu s}/\nu$. Next, $D_x\ell[v] = -\int_0^s D\,\mathrm{div}\,X(\gamma)[Mv]\,d\sigma$, and $\int_0^\infty e^{-\nu\sigma}d\sigma = 1/\nu$. Finally, $D_x^2\ell[v,v'] = -\int_0^s \big(D^2\,\mathrm{div}\,X(\gamma)[Mv, Mv'] + D\,\mathrm{div}\,X(\gamma)[N[v,v']]\big)d\sigma$; bound this with the two previous estimates. □

Differentiation under the integral. Let the integrand be $H(\Phi_{-s}x)w(s,x)$ or $w(s,x)$. By (ODE), and since the flow is defined on a neighbourhood of D, its x-derivatives up to the relevant order are jointly continuous on $[0,\infty)\times D$. By Lemma 8.8 and the identities $D_xw = w\,D_x\ell$ and $D_x^2w = w(D_x^2\ell + D_x\ell\otimes D_x\ell)$, they are dominated by $Ke^{-7s/8}$. Truncating the integral at S, differentiating, and passing to the uniform limit gives the following.

Lemma 8.9. *All constants below depend only on d, uniformly over $\|X - x/d\|_{C^1} \le \varepsilon \le 1/(8d)$.*

(d) *If $X \in C^2$ and $H \in C^1(D)$, then $T_X^{-1}H \in C^1(D)$ and $\|\nabla T_X^{-1}H\|_\infty \le C_d(\|\nabla H\|_\infty + \|H\|_\infty\|D\,\mathrm{div}\,X\|_\infty)$.*

(e) *If $X \in C^3$, then $T_X^{-1}\mathbf{1} \in C^2(D)$, $\|\nabla T_X^{-1}\mathbf{1}\|_\infty \le C_d\|D\,\mathrm{div}\,X\|_\infty$, and $\|D^2T_X^{-1}\mathbf{1}\|_\infty \le C_d\big(\|D^2\,\mathrm{div}\,X\|_\infty + \|D\,\mathrm{div}\,X\|_\infty^2 + \|D\,\mathrm{div}\,X\|_\infty\|D^2X\|_\infty\big)$.*

8.5 The centring function and an exact decomposition

Fix $\varepsilon = \varepsilon_0$ in Lemma 8.4, so that Lemmas 8.5–8.9 apply to $X = V_J$ (with $\varepsilon_0/2$), and on $\mathcal{A}$ also to $X = \hat V$. Let

$$\bar f := T_{V_J}^{-1}\mathbf{1}.$$

Using $\|D^m\,\mathrm{div}\,V_J\|_\infty \le d\|D^{m+1}V_J\|_\infty$, Lemma 8.4(3) and Lemma 8.9(e),

$$\|\bar f\|_\infty \le 2, \qquad \|\nabla\bar f\|_\infty \le CJ2^{J(2-\alpha)_+}, \qquad \|D^2\bar f\|_\infty \le CJ^2 2^{J(3-\alpha)_+}. \tag{7}$$

The last bound uses $2(2-\alpha)_+ \le (3-\alpha)_+$ for $\alpha \ge 1$. Moreover $\bar f \in C^2(D)$, and $\mathrm{div}(\bar f V_J) = 1$ by Lemma 8.7(a′).

Noise term. Let $G_\xi := \operatorname{div}(\bar f \nabla \xi) = \bar f \Delta \xi + \nabla \bar f \cdot \nabla \xi \in C^1(D)$ and $Z := T_{V_J}^{-1} G_\xi$. Since V_J is deterministic, Z is defined for every realisation. By Lemma 8.9(d), $Z \in C^1(D)$, and $T_{V_J} Z = G_\xi$ by Lemma 8.7(a′).

Proposition 8.10. *On $\mathcal{A}$ (so that $\hat V$ satisfies the hypotheses of Lemma 8.5 and $\hat f = T_{\hat V}^{-1}\mathbf{1}$),*

$$\hat f - f = (\bar f - f) - Z + T_{\hat V}^{-1} \operatorname{div}(Z \nabla \xi). \tag{8}$$

Proof. $\hat V$ is C^3 on a neighbourhood of D, so $\hat f \in C^2$ by Lemma 8.9(e), and $T_{\hat V}\hat f = 1$ by Lemma 8.7(a′). Hence $T_{\hat V}(\hat f - \bar f) = 1 - \operatorname{div}(\bar f V_J) - \operatorname{div}(\bar f \nabla \xi) = -G_\xi$. Also $T_{\hat V} Z = T_{V_J} Z + \operatorname{div}(Z\nabla\xi) = G_\xi + \operatorname{div}(Z\nabla\xi)$. Therefore $T_{\hat V}(\hat f - \bar f + Z) = \operatorname{div}(Z\nabla\xi)$, which is continuous. Since $\hat f - \bar f + Z \in C^1$, Lemma 8.7(a) with $X = \hat V$ gives $\hat f - \bar f + Z = T_{\hat V}^{-1} \operatorname{div}(Z\nabla\xi)$, which is (8). □

8.6 The bias: $\|\bar f - f\|_\infty \le C J^2 2^{-J\alpha}$

Proposition 8.11. *Uniformly over $f \in \mathcal{F}(\alpha, r)$, $\|\bar f - f\|_\infty \le C J^2 2^{-J\alpha}$, with no factor J^2 when $\alpha \notin \mathbb{N}$.*

Proof. Reduction. $f \in C^1$ (as $\alpha > 1$), $u \in C^{\alpha+1} \subset C^2$ by (3), $\operatorname{div}(fV) = 1$, and $V_J - V = \nabla e$ on D. So $T_{V_J}(\bar f - f) = 1 - \operatorname{div}(fV_J) = -\operatorname{div}(f\nabla e)$, and by Lemma 8.7(a), $\bar f - f = -T_{V_J}^{-1}\operatorname{div}(f\nabla e)$.

Commutator. Let $g := F - k_J * F$ and $R(x) := -\int k_J(-h)\Delta_h F(x)\Delta_h V^E(x)\,dh$. Then

$$F\,(k_J * V^E) = k_J * (FV^E) + gV^E + R.$$

Indeed, $F(x)(k_J * V^E)(x) - k_J * (FV^E)(x) = \int k_J(y)[F(x) - F(x-y)]V^E(x-y)\,dy$. Insert $F(x) - F(x-y) = -\Delta_{-y}F(x)$ and $V^E(x-y) = V^E(x) + \Delta_{-y}V^E(x)$, and use $\int k_J = 1$; the first part gives gV^E and the second, after $h = -y$, gives R. Let $\Gamma := \operatorname{div}(FV^E) - 1$, which lies in $C^{\alpha-1}(\mathbb{R}^d)$ by the algebra property and vanishes on D. Since $\operatorname{div}(k_J * (FV^E)) = k_J * \Gamma + 1$, on D

$$\operatorname{div}(f\nabla e) = \operatorname{div}(fV_J) - 1 = k_J * \Gamma + \operatorname{div}(gV) + \operatorname{div} R.$$

We treat the three terms in turn, using linearity of $T_{V_J}^{-1}$.

(i) The term $\operatorname{div}(gV)$. Since $V = V_J - \nabla e$, we have $\operatorname{div}(gV) = T_{V_J} g - \operatorname{div}(g\nabla e)$. As $g \in C^1$, Lemma 8.7(a) gives $T_{V_J}^{-1}\operatorname{div}(gV) = g - T_{V_J}^{-1}\operatorname{div}(g\nabla e)$. By (C2) (applied to $F \in C^\alpha$ and $U \in C^{\alpha+1}$): $\|g\|_\infty \le C2^{-J\alpha}$, $\|\nabla g\|_\infty \le C2^{-J(\alpha-1)}$, $\|\nabla e\|_\infty \le C2^{-J\alpha}$ and $\|D^2 e\|_\infty \le C2^{-J(\alpha-1)}$. By Lemma 8.7(b), the contribution is $\le C2^{-J\alpha} + C2^{-J(2\alpha-1)}$.

(ii) The term $\operatorname{div} R$. We have $\partial R = -\int k_J(-h)\big[\Delta_h \partial F\,\Delta_h V^E + \Delta_h F\,\Delta_h \partial V^E\big]dh$. Each product has the form $\Delta_h\phi_1\,\Delta_h\phi_2$ with $\phi_i \in C^{s_i}$ and $(s_1, s_2) = (\alpha - 1, \alpha)$ or $(\alpha, \alpha - 1)$. By (C3), and using $|P_{\phi_i}| \le C|h|$ for $|h| \le 1$, the product equals a polynomial in h (with coefficients depending on x) whose monomials have degrees in $[2, s_1 + s_2) = [2, 2\alpha - 1)$. The remainder is bounded by $C|h|^{\min(1+s_1,\,1+s_2,\,s_1+s_2)}\Lambda(h)^2 = C|h|^\alpha\Lambda(h)^2$. When $s_i < 1$, P_{ϕ_i} is empty. The polynomial terms integrate to 0 against $k_J(-h)$, because $N_0 \ge 2\alpha + 2$. Since k_J is supported in $|h| \le C2^{-J}$, the substitution $h = 2^{-J}y$ gives $\int |k_J(h)||h|^\alpha\Lambda(h)^2 dh \le CJ^2 2^{-J\alpha}$. So $\|\operatorname{div} R\|_\infty \le CJ^2 2^{-J\alpha}$, with no J^2 for non-integer α (then no logarithm appears in (C3)). By Lemma 8.7(b), the contribution is $\le CJ^2 2^{-J\alpha}$.

(iii) The term $k_J * \Gamma$. On D, $k_J * \Gamma = k_J * \Gamma - \Gamma$, so by (C2) with $s = \alpha - 1$ and $m = 0$ it is $O(2^{-J(\alpha-1)})$. By the support of k_J, it vanishes at points of D at distance $> C2^{-J}$ from ∂D. By Lemma 8.7(c) with $\delta = C2^{-J}$, the contribution is $\le C2^{-J}2^{-J(\alpha-1)} = C2^{-J\alpha}$.

Adding (i)–(iii) gives the claim. □

8.7 The variance: $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$

Proposition 8.12. $\mathbb{E}\|Z\|_{L^2(D)}^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$.

Proof. Representation. Let $\mathcal{D}_y := \bar f(y)\Delta_y + \nabla\bar f(y)\cdot\nabla_y$. By (G), $G_\xi(y) = n^{-1/2}\mathbb{W}(\mathcal{D}_y k_J(y,\cdot))$, and since Bochner integrals commute with $\mathbb{W}$,

$$Z(x) = n^{-1/2}\mathbb{W}(h_x), \qquad h_x = \int_0^\infty w(s,x)\,\mathcal{D}_y k_J(y,\cdot)\big|_{y=\gamma_x(s)}\,ds,$$

where γ_x is the backward ray of V_J. Hence $\mathbb{E}Z(x)^2 = n^{-1}\|h_x\|_2^2$.

Kernel bound. Since $(2-\alpha)_+ < 1$, (7) gives $\|\nabla\bar f\|_\infty \le CJ2^{J(2-\alpha)_+} \le C2^J$. With (K1) and (K2), $|\mathcal{D}_y k_J(y,z)| \le C2^{J(d+2)}\mathbf{1}\{|y-z|\le\delta_J\}$, where $\delta_J := C_0 2^{-J}$. Let $\mu_x := \int_0^\infty w(s,x)\,\delta_{\gamma_x(s)}\,ds$, a measure on D of mass $\le 8/7$. Then

$$|h_x(z)| \le C2^{J(d+2)}\mu_x(B(z,\delta_J)), \qquad \|h_x\|_2^2 \le C2^{2J(d+2)}\int \mu_x(B(z,\delta_J))^2 dz \le C2^{2J(d+2)}\delta_J^d\int \mu_x(B(y,2\delta_J))\,\mu_x(dy).$$

The last step is Fubini: $\int\mu(B(z,\delta))^2dz = \iint |B(y,\delta)\cap B(y',\delta)|\,\mu(dy)\mu(dy')$, and the intersection is empty unless $|y-y'|\le 2\delta$.

Ray measure. Let x_J be the zero of V_J, $\rho = |\cdot - x_J|$, and $\rho_x = \rho(x) > 0$. Let $\varrho(s) = \rho(\gamma_x(s))$. By Lemma 8.5(4),(6), $w(s,x) \le (\varrho(s)/\rho_x)^\beta$ with $\beta = d(1-d\varepsilon)/(1+d\varepsilon) \ge 7d/9 > 1$. Moreover ϱ decreases strictly from ρ_x to 0, with $|\varrho'| \ge \nu\varrho$. Changing variables $t = \varrho(s)$, for every $\varphi \ge 0$,

$$\int \varphi\,d\mu_x \le \frac{1}{\nu}\int_0^{\rho_x} \varphi(\gamma_x \text{ at radius } t)\,(t/\rho_x)^\beta\,\frac{dt}{t}. \tag{9}$$

Small-ball mass. If $|\gamma_x(s)-y| \le 2\delta_J$, then $|\varrho(s)-\rho(y)| \le 2\delta_J$. Hence, by (9), $m(y) := \mu_x(B(y,2\delta_J))$ satisfies

$$m(y) \le C\delta_J\rho(y)^{\beta-1}\rho_x^{-\beta} \text{ if } \rho(y)\ge 4\delta_J, \qquad m(y) \le C\min(1,\delta_J/\rho_x)^\beta \text{ if } \rho(y) < 4\delta_J.$$

(In the first case $t \in [\rho(y)/2, 3\rho(y)/2]$ ranges over an interval of length $4\delta_J$; in the second $t \le \min(6\delta_J,\rho_x)$.)

Integration. Apply (9) once more, with $\varphi = m$. The part with $t \ge 4\delta_J$ is $\le C\delta_J\rho_x^{-2\beta}\int_0^{\rho_x} t^{2\beta-2}dt \le C\delta_J/\rho_x$, since $2\beta-1>0$. The part with $t < 4\delta_J$ is $\le C\min(1,\delta_J/\rho_x)^{2\beta} \le C\min(1,\delta_J/\rho_x)$. Trivially, $\int m\,d\mu_x \le \mu_x(D)^2 \le 4$. Hence $\int m\,d\mu_x \le C\min(1,\delta_J/\rho_x)$, and

$$\|h_x\|_2^2 \le C2^{J(d+4)}\min(1,2^{-J}/\rho_x) \le C2^{J(d+3)}|x-x_J|^{-1}.$$

Since $\int_D |x-x_J|^{-1}dx \le C$ for $d\ge 2$, Tonelli's theorem and (4) give $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$. □

8.8 The remainder and the bad event

Lemma 8.13. *On $\mathcal{A}$, $\|T_{\hat V}^{-1}\operatorname{div}(Z\nabla\xi)\|_\infty \le CJ^4 2^{-J(2\alpha-1)}$. This is $o(2^{-J\alpha})$ because $\alpha > 1$.*

Proof. We use (5), (7), $(2-\alpha)_+ < 1$ and $(3-\alpha)_+ \le 2$. First,

$$\|G_\xi\|_\infty \le \|\bar f\|_\infty\|\Delta\xi\|_\infty + \|\nabla\bar f\|_\infty\|\nabla\xi\|_\infty \le CJ2^{-J(\alpha-1/2)} + CJ^2 2^{J(2-\alpha)_+ - J(\alpha+1/2)} \le CJ^2 2^{-J(\alpha-1/2)}.$$

Next, $\|\nabla G_\xi\|_\infty \le C(\|D^3\xi\|_\infty + \|\nabla\bar f\|_\infty\|D^2\xi\|_\infty + \|D^2\bar f\|_\infty\|\nabla\xi\|_\infty) \le CJ^3 2^{-J(\alpha-3/2)}$. Here each cross term has exponent $\le -(\alpha-3/2)$ because $\alpha \ge 1$.

By Lemma 8.7(b) and Lemma 8.9(d), with $\|D\operatorname{div} V_J\|_\infty \le d\|D^2 V_J\|_\infty \le CJ2^{J(2-\alpha)_+}$ from Lemma 8.4(3), we get $\|Z\|_\infty \le CJ^2 2^{-J(\alpha-1/2)}$ and $\|\nabla Z\|_\infty \le C(\|\nabla G_\xi\|_\infty + \|G_\xi\|_\infty J2^{J(2-\alpha)_+}) \le CJ^3 2^{-J(\alpha-3/2)}$. Hence

$$\|\operatorname{div}(Z\nabla\xi)\|_\infty \le \|\nabla Z\|_\infty\|\nabla\xi\|_\infty + \|Z\|_\infty\|\Delta\xi\|_\infty \le CJ^4 2^{-J(2\alpha-1)} + CJ^3 2^{-J(2\alpha-1)} \le CJ^4 2^{-J(2\alpha-1)}.$$

Lemma 8.7(b) for $X = \hat{V}$ concludes the proof. □

Lemma 8.14. *$P(\mathcal{A}^c) \le Ce^{-cJ^2}$, uniformly in f.*

Proof. Each component $\partial^\iota\xi(x) = n^{-1/2}\mathbb{W}(\partial_x^\iota k_J(x,\cdot))$, $|\iota| = m$, is centred Gaussian. By (K3) its standard deviation is $\le C\bar\sigma_m$, where $\bar\sigma_m := n^{-1/2}2^{J(d/2+m)}$, and its canonical metric is $\le C\bar\sigma_m 2^J|x-x'|$. The covering numbers of D in this metric are therefore $\le (C\bar\sigma_m 2^J/\eta)^d$, and Dudley's bound gives $\mathbb{E}\sup_D|\partial^\iota\xi| \le C\bar\sigma_m\sqrt{J} \le J\bar\sigma_m/2$ for large J. The Borell–TIS inequality at deviation $J\bar\sigma_m/2$ gives $P(\sup_D|\partial^\iota\xi| > J\bar\sigma_m) \le 2e^{-cJ^2}$. A union bound over the finitely many ι with $1 \le |\iota| \le 3$ concludes the proof. □

8.9 Assembly

Proof of Theorem 4.1(ii). Let $r \le r_0$ (the minimum of the values required in Lemmas 8.1 and 8.4) and $J \ge J_0$. On $\mathcal{A}$ we have $\mathcal{A} \subset \mathcal{B}$, so (8) applies. Clipping only decreases $|\tilde f - f|$, so Proposition 8.11 and Lemma 8.13 give

$$\|\tilde f - f\|_2^2 \le 3|D|\big(CJ^2 2^{-J\alpha}\big)^2 + 3\|Z\|_2^2 + 3|D|\big(CJ^4 2^{-J(2\alpha-1)}\big)^2.$$

Off $\mathcal{A}$, $|\tilde f - f| \le 2r$ pointwise, so $\|\tilde f - f\|_2^2 \le 4r^2|D|$. Taking expectations, using Proposition 8.12, Lemma 8.14 and (4),

$$\mathbb{E}_f\|\tilde f - f\|_2^2 \le CJ^4 2^{-2J\alpha} + C2^{-2J\alpha} + CJ^8 2^{-2J(2\alpha-1)} + Ce^{-cJ^2} \le CJ^4 2^{-2J\alpha} \le C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)},$$

uniformly over $\mathcal{F}(\alpha, r)$. The finitely many n with $J < J_0$ are covered by the trivial bound $4r^2|D|$. For non-integer α, Lemma 8.3 and (C3) produce no logarithm. Then the bias is $\le C2^{-J\alpha}$, and the remaining powers of J multiply terms of strictly smaller order. So the bound is $Cn^{-2\alpha/(2\alpha+d+3)}$. □

Theorem 4.1 follows from Sections 7 and 8.

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